

Global transport in soaring flights

This deck develops the whole chapter without a prescribed duration. Its central distinction is between measured constraints on the observed ground trajectories and hypotheses about their generating process.

Version: 14/09/2026

Chapter 3

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f = |\mathbf{X}_f|.$$

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \underbrace{\frac{1}{n_f(\tau)} \sum_k R_{fk}^2}_{\text{mean within flight } f}.$$

Windows stay within a retained segment. The main comparison covers $10 \leq \tau \leq 10^4$ s.

Displacement growth, distribution shape, directional preference and memory must be confronted together.

Archive TAMSD: equal segment weights. Full V_1, V_2 : equal flight weights. Baseline quantiles and moments: pooled windows.

What motion should a stochastic model reproduce?

The full diagnostic collector has no flight cap and no longest-segment-only restriction. Flights recorded more slowly than ten seconds are ineligible for its common grid. Each lag uses every supported flight; a separate fixed long-flight comparison controls population change. Different grids, minimum segment sizes and weights explain some differences from the archive TAMSD.

Estimator	Unit receiving equal weight	Support requirement
Launch ensemble	available flights at elapsed time t	observed coordinate at that time
Archive TAMSD	eligible retained segments	native-grid displacement within segment
Full V_1, V_2	eligible flights at lag τ	within-segment common-grid stencil
Pooled moments	available increment origins	stated non-overlapping baseline stencil

Different exponents can reflect different weights and changing support. Their difference alone is not an ergodicity test.

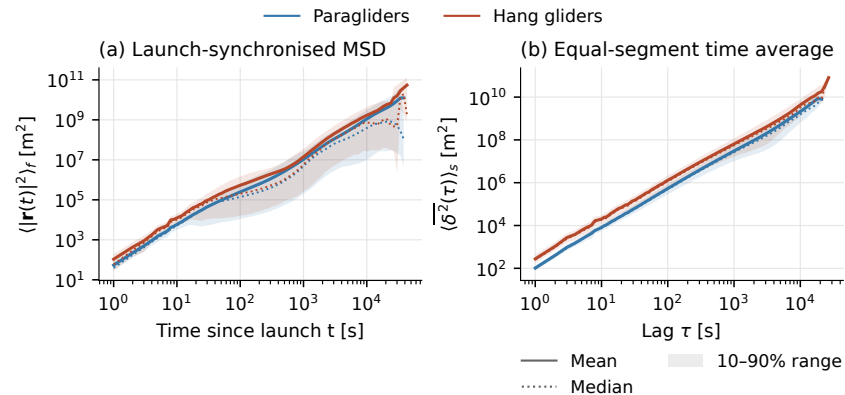
Which estimator corresponds to the physical population the model should reproduce?

Launch, segment and flight averages use different weights

Within a flight, contributions are combined using their admissible origins. The archive TAMSD and the full diagnostic reporter use different grids and minimum segment sizes, so their flight counts need not be identical. State the estimator before comparing numerical slopes.

Launch averages and time averages use different populations

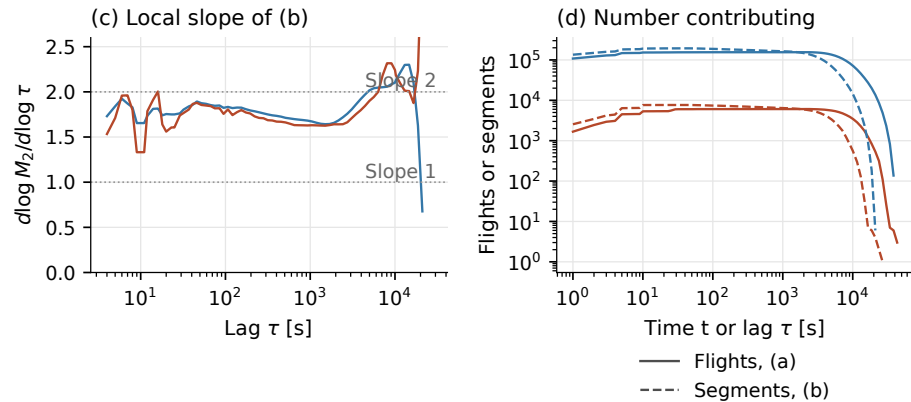
These are the upper two panels of the complete archive MSD figure. Blue denotes paragliders and terracotta hang gliders. The quantities use different clocks and weighting; their comparison is not an ergodicity test.



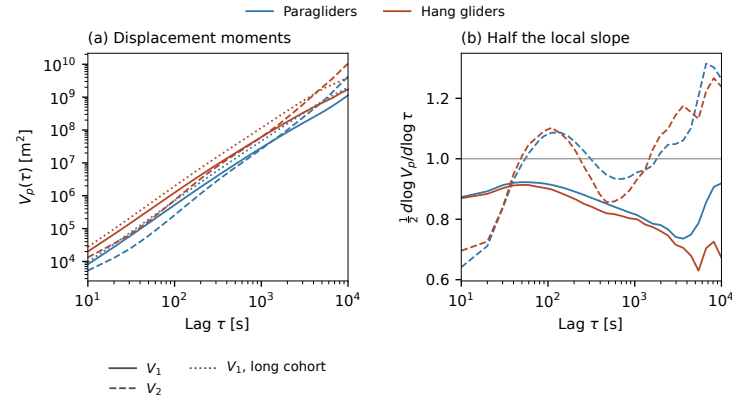
How much of the difference follows launch synchronization and segment weighting?

Local slopes must be read together with falling support

The lower panels of the archive MSD figure show local slopes and contributing flights or segments. A small bootstrap error does not remove scale dependence or population selection.



Which scale interval supports a stable finite-range description?



Full eligible archive; 10–10,000 s. Solid: V_1 ; dashed: V_2 ; dotted: V_1 , fixed long cohort. Right: half the local slope. Descriptive fits; no calibrated confidence intervals.

The MSD grows faster than linearly on the measured interval

The displayed exponents are imported from the completed full report. The first- and second-difference estimators use equal flight weights and do not cross segment boundaries. Their local slopes bend, so these are descriptive fits over the stated range. They differ from the equal-segment archive TAMSD and from pooled-origin high-order moments.

	b_1	b_2
Para	1.703	2.003
Hang	1.649	1.985

$V_1 = \langle R^2 \rangle$; V_2 uses second differences.

The local slope bends. One fitted number does not establish a power law.

$$A_2 \mathbf{r} = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad V_2 = \langle |A_2 \mathbf{r}|^2 \rangle.$$

Exact cancellation

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_d t + \mathbf{Y}(t) \quad \Rightarrow \quad A_2 \mathbf{r} = A_2 \mathbf{Y}.$$

The constant velocity may combine wind, route choice and net progress. A turn or spatially varying wind survives.

Conditional interpretation

If centred increments have finite variance, stationary statistics and the law $K\tau^{2H}$,

$$V_2 = K(4 - 2^{2H})\tau^{2H}, \quad 0 < H < 1.$$

At $H = 1$ the prefactor vanishes. Our $b_2 \simeq 2$ cannot be read as a validated $H \simeq 1$.

Which residual and which scale interval could satisfy the assumptions needed to estimate H ?

At $\tau = 10^4$ s, V_2 retains only 14,360 para / 563 hang flights; it needs 2τ of continuous support.

Constant velocity cancels; an intrinsic H remains open

Invariant means unchanged after adding a constant position offset and a constant velocity times time. It does not mean constant as τ changes. For adjacent increments I_1 and I_2 , their sum spans 2τ and their difference is A_2 . The parallelogram identity gives $V_2 = 4M(\tau) - M(2\tau)$ for the same stationary-increment population. Substituting $M(\tau) = K\tau^{2H}$ to the power $2H$ gives the displayed formula. All three conditions matter: finite variance, stationary increment statistics, and the same power law through 2τ . Stationarity means that the increment law at a given duration does not change with the start time; it does not require independence. A fitted b_2 above two does not by itself identify which assumption fails. A random-phase circular trajectory has stationary increments and finite variance but no exact increment power law, and V_2 grows as τ to the fourth at small lag.

$$\bar{\mathbf{v}}_1 = \frac{\mathbf{r}(t + \tau) - \mathbf{r}(t)}{\tau}, \quad \bar{\mathbf{v}}_2 = \frac{\mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau)}{\tau}.$$

$$\frac{A_2 \mathbf{r}}{\tau} = \bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1, \quad D_2(\tau) = \frac{\sqrt{V_2(\tau)}}{\tau} = \sqrt{\langle |\bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1|^2 \rangle}.$$

D_2 is in m/s. At $\tau = 60$ s it compares the mean velocity vector in one minute with the vector in the next minute. Changes of magnitude and direction both contribute.

At which scales does the ground motion change most, and how does that depend on the region?

Exact kinematic identity; no power law or stationarity assumption is needed.

Second differences measure changes of interval-mean velocity

This is an RMS change of interval-mean ground velocity. It is neither instantaneous acceleration nor wind speed. A constant added velocity cancels in both intervals. Keeping the physical units makes regional amplitudes interpretable even when a single exponent is not supported. The later regional study uses this quantity together with the vector covariance, distributions and order-three control.

$$V_2(\tau) = \tau^2 D_2^2(\tau), \quad b_{2,\text{loc}} = 2 + 2 \frac{d \log D_2}{d \log \tau}.$$

Saved full-archive curves

Over 10–10,000 s, D_2 varies within:

	D_2 range (m/s)
Paragliders	4.9–7.3
Hang gliders	7.6–11.5

A nonmonotone change of less than a factor two across three decades of lag.

Interpretation and limits

The τ^2 factor can dominate growth; local curvature prevents one validated H .

The fixed long-flight cohort also gives $b_2 > 2$: flight membership alone cannot explain it. Origins and weights still change.

Does a stable slope survive with flights and origins fixed, within comparable flight phases?

Why a slope near two need not identify a Hurst exponent

D_2 is the RMS difference between mean ground-velocity vectors in adjacent intervals, with the same per-lag averaging as V_2 . These ranges were checked directly from the saved full-archive arrays; no flight subsampling or new transport fit was used. Approximate constancy explains how a descriptive slope near two arises, but does not identify its physical cause. At the exact $H=1$ boundary of the stationary-increment power-law model, the second difference would have zero variance. Small fitted excesses above two have no calibrated significance here. Keeping only long flights fixes membership but not the contributing origins or their weights. Cadence and smoothing sensitivity should accompany a search for a restricted stable regime; the current evidence does not rule out every useful finite-range H .

A stationary centred velocity with variance σ_v^2 and normalised correlation $C(s) = e^{-s/t_p}$ gives

$$V_2(\tau) = 2\sigma_v^2 t_p^2 \left[2\frac{\tau}{t_p} - 3 + 4e^{-\tau/t_p} - e^{-2\tau/t_p} \right].$$

$$\tau \ll t_p : V_2 \sim \frac{4\sigma_v^2}{3t_p} \tau^3, \quad \tau \gg t_p : V_2 \sim 4\sigma_v^2 t_p \tau.$$

A local slope near two occurs between these regimes despite stationary position increments. [This is an analytical example, not a fit to the flights.](#)

In flights, turns, glide–climb transitions and changing wind survive constant-velocity cancellation. Their characteristic times, combined with changing observation support, are a plausible origin of the measured curvature.

Which dominates the bends: turning, phase changes or sampling? Can phase-conditioned and common-origin comparisons separate them?

Thesis Sec. 3.1.2 and the velocity-covariance model in Sec. 3.6; the closed formula was checked against covariance integration.

Matteo Di Sante | Chapter 3: global transport

Finite persistence can produce a crossover without nonstationarity

For the same stationary-increment population, $V_2(\tau) = 4M_2(\tau) - M_2(2\tau)$. A 2H interpretation requires one increment power law at both τ and 2τ , not merely stationarity. Exponential velocity covariance introduces a persistence time and gives a cubic-to-linear crossover; a differentiable curved path can instead give a quartic small-lag limit. Neither limit is claimed to describe all measured lags. The observed D2 curves are nonmonotone and need not follow one exponential model. Anisotropy alone does not invalidate the vector variance identity. We have a plausible explanation in terms of finite physical scales and sampling, not an identified wind effect or demonstrated nonstationarity. When conditioning on phases, retain an explicit category for windows crossing transitions.

$$A_3 \mathbf{r} = \mathbf{r}(t + 3\tau) - 3\mathbf{r}(t + 2\tau) + 3\mathbf{r}(t + \tau) - \mathbf{r}(t), \quad V_3 = \langle |A_3 \mathbf{r}|^2 \rangle.$$

For $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$ with constant $\mathbf{a}$,

$$V_2 = |\mathbf{a}|^2 \tau^4, \quad V_3 = 0.$$

Under the centred stationary-increment power law,

$$\frac{V_3}{V_2} = \frac{15 - 6 \cdot 2^{2H} + 3^{2H}}{4 - 2^{2H}}, \quad 0 < H < 1.$$

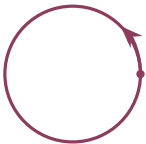
Do the same supported observations satisfy the predicted relation between orders?

Order three requires 3τ of continuous support; the model law must hold through 3τ .

Third differences also cancel a constant acceleration

V_3 is a sensitivity check to quadratic position trends. A smooth turn survives: typically V_2 scales as τ to the fourth and V_3 as τ to the sixth at small lag. Under the stated model both orders have exponent $2H$ but different amplitudes; Brownian motion plus constant velocity gives $V_3/V_2=3$. Independent position error with variance σ^2 per coordinate contributes $6\sigma^2$ at order two and $20\sigma^2$ at order three. Correlated errors and smoothing require a different noise calculation. Agreement between orders does not prove that environmental forcing has been removed.

Open straight course



Closed circular course

Full eligible archive

	Task	<i>N</i>	<i>b</i> ₁	<i>b</i> ₂
Para	Open	67003	1.770	1.937
	Closed	87807	1.623	2.036
Hang	Open	2183	1.688	1.892
	Closed	3877	1.628	2.007

The ordering reverses at order two. Constant-drift cancellation leaves the return geometry in the observable.

Compare tasks at matched duration and region, using both physical lag and τ/T .

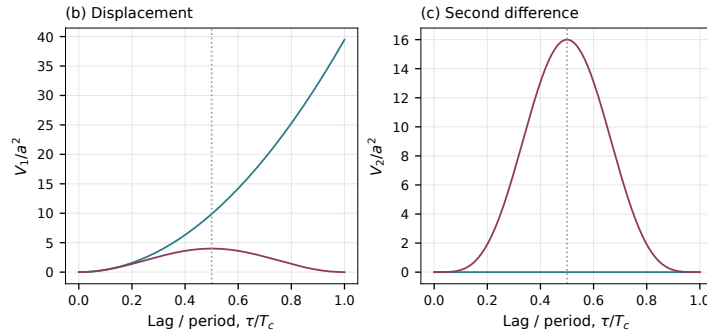
Task declarations; descriptive slopes, 10–10,000 s. Drawing is a geometric illustration. Exact circular example.

Open and closed tasks change the growth of displacements

The task declaration supplies open/closed labels; it does not force every retained segment to return to its start. Counts and slopes are read from the full-run report. Match duration and environment before interpreting task contrasts. The exact circular example illustrates how geometry survives second differencing; it is not a fitted flight model.

For $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$ and $T_c = 2\pi/\omega$,

$$V_1 = 4a^2 \sin^2(\pi\tau/T_c), \quad V_2 = 16a^2 \sin^4(\pi\tau/T_c).$$



Which scale effects could follow return geometry without a change of stochastic law?

A circle retains its geometry after second differencing

This is an exact periodically continued circle, not a fitted flight model. Constant speed does not imply ballistic endpoint displacement over every lag. The original two variation panels are retained; the circular path was illustrated on the preceding task slide.

$$c_f^{\text{obs}} = \frac{|\mathbf{r}_{f,\text{last}} - \mathbf{r}_{f,\text{first}}|}{\sum_{s \in \mathcal{S}_f} \sum_k |\mathbf{r}_{fs,k+1} - \mathbf{r}_{fs,k}|}.$$

The denominator omits unobserved intervals. For a segmented flight, c_f^{obs} need not be bounded by one.

Declared task, observed endpoint separation and actual flown-route closure are related descriptions with different information requirements.

Does the task contrast persist when restricted to continuously observed flights?

Observed-path closure needs special care for segmented flights

The descriptor preserves recorded endpoints without inventing missing path length. Reacquisition offsets can affect it even though every increment used for transport remains inside one segment. Use the task labels and this observed-path ratio as distinct descriptive stratifications; neither is an independent route-quality label.

$$Q_p(\tau) = \inf\{r : F_\tau(r) \geq p\}.$$

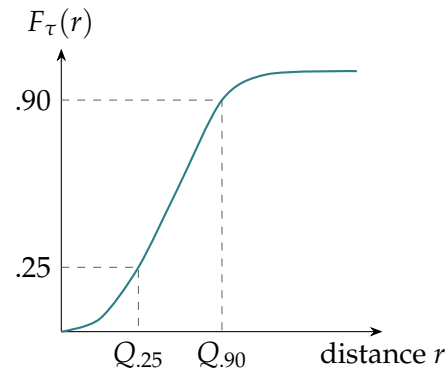
An H -self-similar process with stationary increments predicts

$$R(\tau) \stackrel{d}{=} (\tau/\tau_0)^H R(\tau_0),$$

$$Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0).$$

$$H_p = H \text{ for every } p; \quad Q_{.90}/Q_{.25} \text{ stays fixed.}$$

Schematic CDF. Quantiles describe ranks within one distribution; they do not define permanent classes of slow and fast flights. Equal slopes are necessary, not sufficient, for process self-similarity.



A single displacement scale predicts equal quantile slopes

The lower quartile probes relatively small displacements, while the 90th percentile probes larger ones. Both use the same distribution at each lag. A different flight may supply an observation near each rank, and rank identities may change; that does not invalidate a quantile. The mathematical issue is whether the population and weighting used to construct the distribution change with lag. No isotropy assumption is needed for the radial scaling implication.

Population	Paragliders	Hang gliders
Retained cleaned flights	156,406	6,094
Eligible for the common-grid diagnostic	155,085	6,060
Fixed long-flight control	14,360	563

Every eligible segment contributes where it supports the lag. The long-flight control adds a common population and common origins across the complete lag range.

Which statement concerns the observed archive, and which is conditional on long flights?

The full archive and the fixed cohort answer different questions

The common ten-second grid requires native cadence at most ten seconds and sufficient within-segment support. These are eligibility conditions, not flight-number caps imposed by the Mac. The twenty-thousand-second criterion is an operational choice for the additional control. It is not mathematically necessary merely to observe one increment at a ten-thousand-second lag.

All eligible flights contribute at each supported lag in the archive-wide analysis. The additional fixed cohort contains every flight with a retained segment lasting at least 20,000 s.

This provides at least 1001 common ten-second origins in one qualifying segment at the largest lag of 10,000 s. Shorter segments of those flights contribute wherever they support the same full lag stencil.

How different are the shorter-flight results on a shorter common lag interval?

The fixed control requires common temporal support

The 20,000-second threshold is more restrictive than mere existence of a 10,000-second increment. It is an explicit support convention for this sensitivity analysis. The need for long observations is mathematical; the particular factor of two and chosen maximum lag are design choices, not necessities imposed by memory limits.

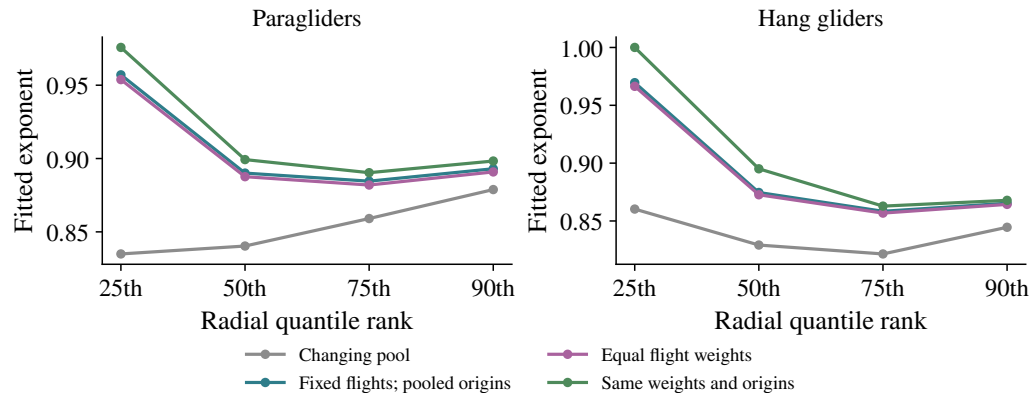
1. All available flights and origins, pooled.
2. Fixed flights, still pooling their available origins.
3. Fixed flights with equal total flight weight.
4. Fixed flights, equal flight weight and the same origins at every lag.

$$F_{\tau}(r) = \frac{1}{N_*} \sum_f \frac{1}{n_f^*} \sum_{k \in \mathcal{O}_f^*} \mathbf{1}\{R_{fk}(\tau) \leq r\}.$$

How much of a change in the quantile curve comes from membership, weights or origin selection?

Holding flights fixed does not by itself hold the observations fixed

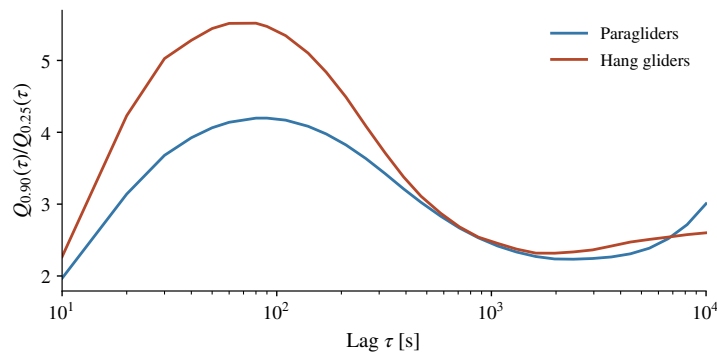
All common origins stay within a retained segment for every requested lag. Overlapping increments are deliberately retained; they are not independent replicates. The final control uses 14360 paraglider flights with 21739115 common origins and 563 hang-glider flights. Read the final hang-glider origin count from its completed report rather than a preflight estimate.



Which part of the slope change comes from each population control?

Membership, flight weights and common origins change the comparison

The four controls use the same ranks and fitting interval. The fixed cohort includes every qualifying long flight. Curves here are descriptive point estimates; the final fixed-control uncertainty uses the whole-flight paired bootstrap.



$$\frac{d \log(Q_{.90}/Q_{.25})}{d \log \tau} \\ = H_{.90}(\tau) - H_{.25}(\tau).$$

The ratio first rises, then falls, then rises again.

A negative fitted slope summarizes part of that evolution; it does not imply narrowing at every lag.

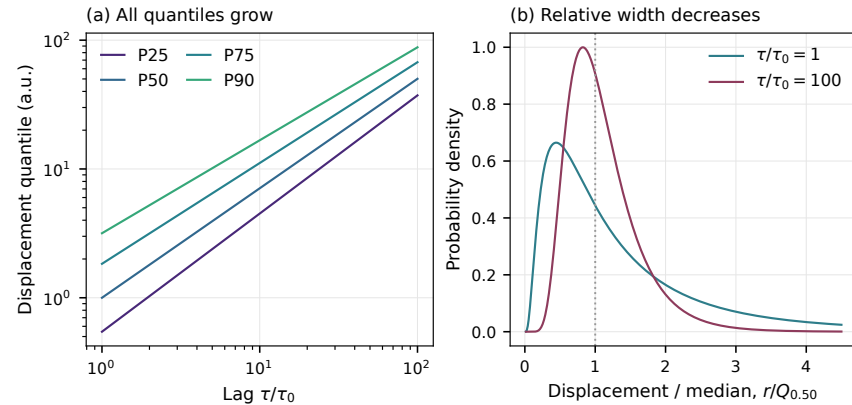
How much of this shape evolution survives site–day uncertainty, route matching and phase conditioning?

Fully controlled population. A lower ratio means less relative spread by this measure; it does not establish a flatter density tail.

Relative spread changes, but the change is not monotone

The fixed-control ratio starts near 1.97 for paragliders and 2.27 for hang gliders, peaks near ninety and eighty seconds, and has a post-peak minimum near two thousand seconds. The final values near 3.01 and 2.60 exceed the initial ones. A negative fitted log-ratio slope is therefore not monotone narrowing.

All quantiles can grow while spread relative to the median narrows



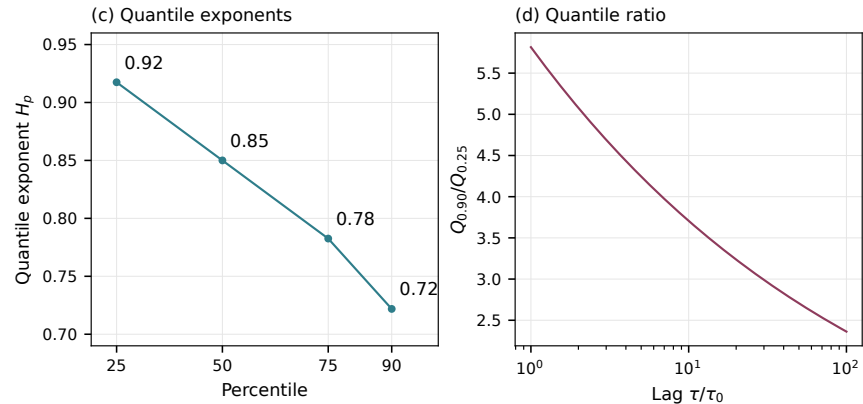
How can a changing distribution shape coexist with increasing displacement scales?

All quantiles can grow while spread relative to the median narrows

Analytical illustration: $R(\tau) = (\tau/\tau_0)^{0.85} \exp[\sigma(\tau)Z]$, with standard Gaussian Z and $\sigma(\tau) = 0.9 - 0.1 \log(\tau/\tau_0)$ on the illustrated range. This is not a fit to flight data.

Unequal rank exponents can encode that changing shape

For the same lognormal illustration, $H_p=0.85-0.1 \Phi^{-1}(p)$. The example explains the algebra without assigning permanent slow/fast identities to particular flights. The actual empirical curves need their own population controls and uncertainty estimates.



Which empirical observations would distinguish this example from a true common scale family?

$$\mathbf{X}(\tau) \stackrel{d}{=} \tau^H \mathbf{Z} \implies |\mathbf{X}(\tau)| \stackrel{d}{=} \tau^H |\mathbf{Z}|,$$

even if the law of $\mathbf{Z}$ is anisotropic.

With two centred independent Gaussian components instead,

$$\mathbb{E}[R^2] = a\tau^{2H_E} + b\tau^{2H_N},$$

unequal component exponents can produce a lag-dependent radial effective exponent.

Is a mismatch caused by unequal directional scales, changing marginal shape or changing joint structure?

Anisotropy does not prevent a common scalar rescaling

Scalar self-similarity and isotropy are separate hypotheses. Operator self-similarity allows a matrix exponent and is broader, but does not permit arbitrary lag-by-lag normalization chosen to force collapse. The current plots assess the stated scalar comparison; they do not exhaust that wider model class.

$\{(1,2), (-1,-2)\}$ versus $\{(1,-2), (-1,2)\},$

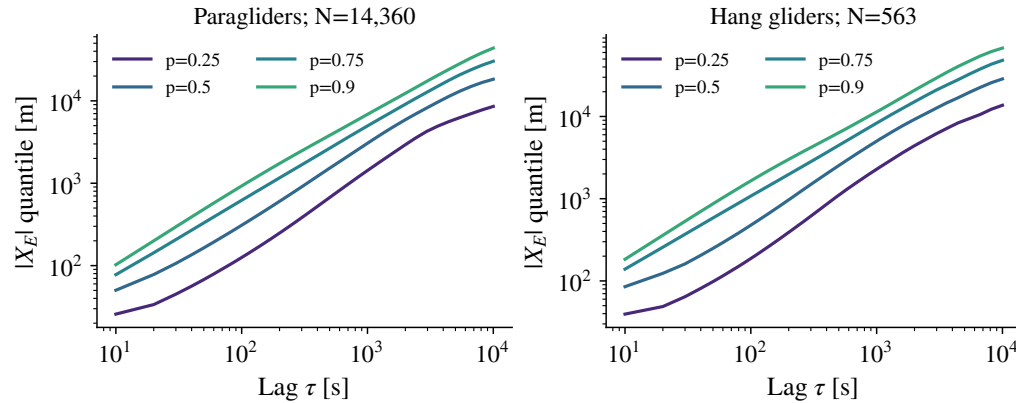
each pair equiprobable.

Quantity	First law	Second law
Signed east marginal	± 1	± 1
Signed north marginal	± 2	± 2
Radius	$\sqrt{5}$	$\sqrt{5}$
Cross-covariance	$+2$	-2

The missing information is real: signs and component dependence require a joint diagnostic.

East, north and radius together still do not identify the joint law

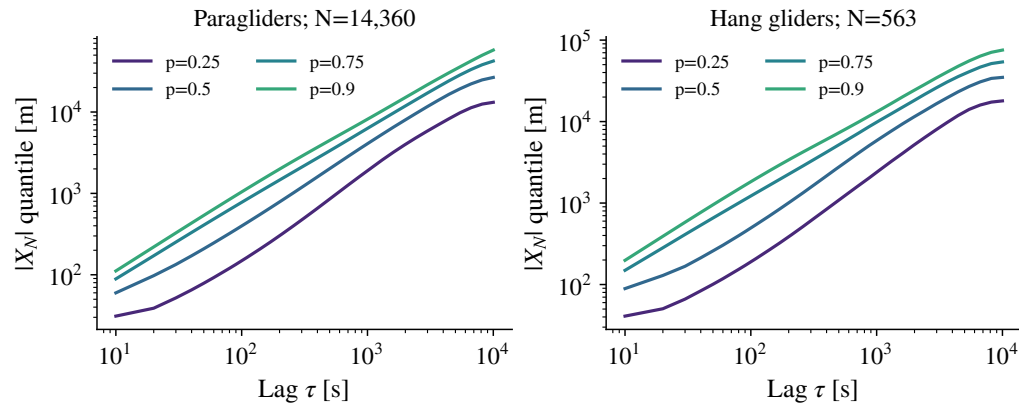
This is an exact mathematical counterexample, not a simulated flight result. Even signed marginals and radius can agree while the joint laws differ. The Cramer–Wold characterization uses all signed linear projections, not only the two coordinate projections. The signed joint plots address additional structure at their stated bin resolution.



Do the ranks retain one relative shape as the lag increases?

Fixed-population quantiles: east component

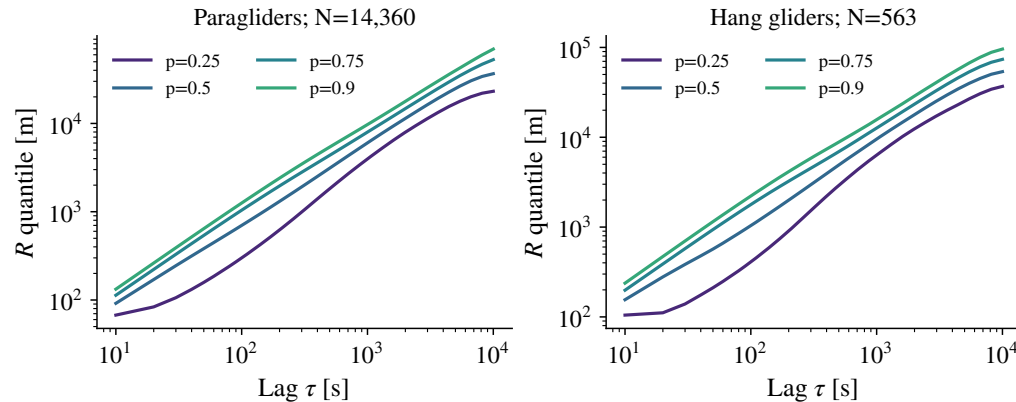
Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.



Do the ranks retain one relative shape as the lag increases?

Fixed-population quantiles: north component

Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.



Do the ranks retain one relative shape as the lag increases?

Fixed-population quantiles: radial displacement

Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.

Resample flights with replacement 400 times. Each selected flight retains all its common origins and its joint east–north observations at every lag.

Overlapping origins remain within the resampled flight; they are not treated as independent replicates.

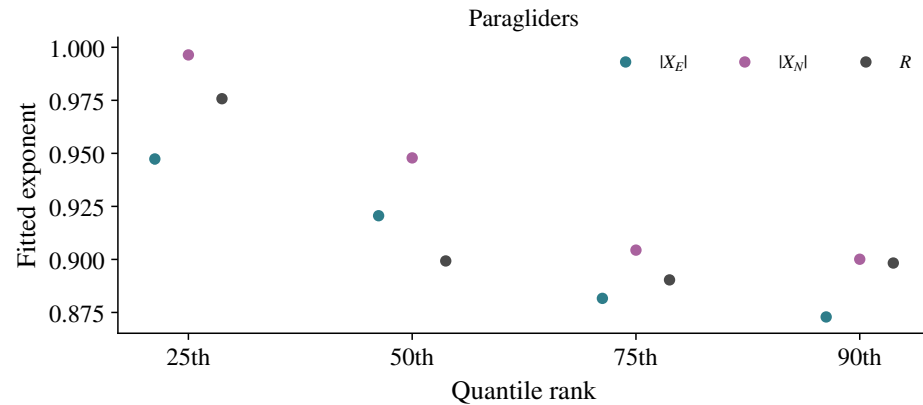
How would shared pilot history or site–day weather change these uncertainty estimates?

Whole-flight bootstrap preserves pairing across ranks, axes and lags

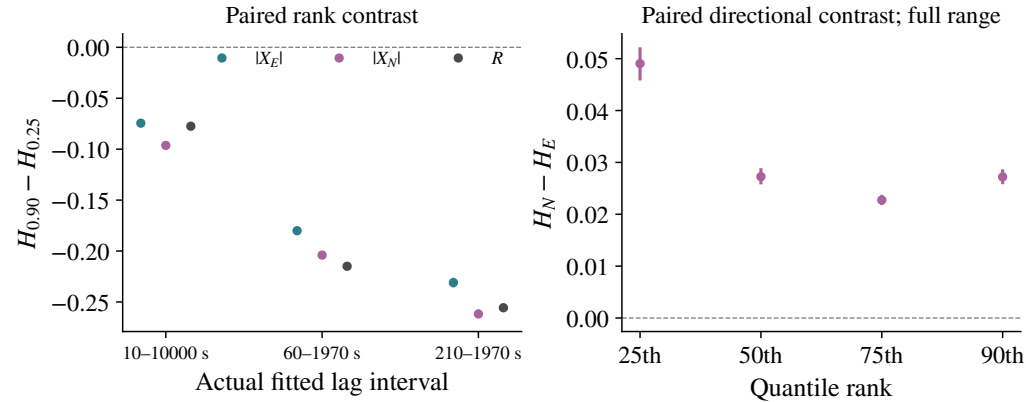
The current intervals assume independent flights within the fixed long-flight population. They are not iid increment intervals, simultaneous bands over all comparisons, or measurement-error intervals. Paired differences are computed inside each replicate. A large complete archive does not remove the assumptions behind the resampling model.

Paragliders: component and radial exponent estimates

Markers are full-range fitted exponents; vertical segments are the exact saved 95-percent percentile endpoints from 400 whole-flight bootstrap draws. They are not simultaneous intervals and do not include sensor, site-day or fitting-range uncertainty.



Which differences need a paired contrast instead of comparing overlapping intervals?



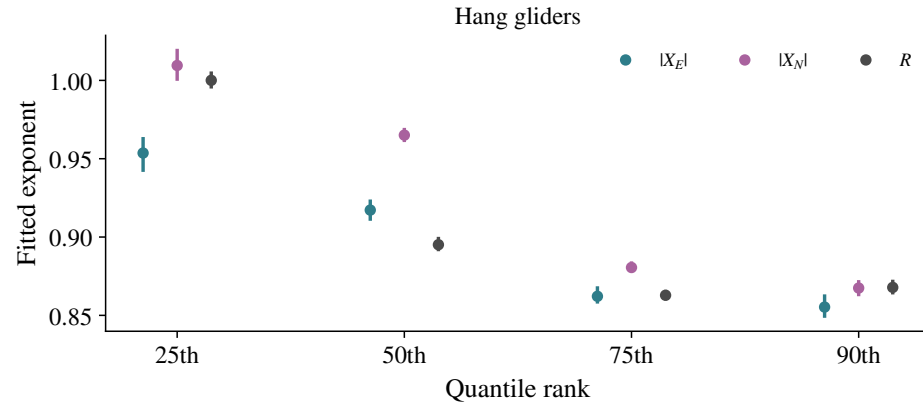
Which differences persist under a change of fitting interval?

Paragliders: paired rank and directional contrasts

The full-range north-minus-east contrast is positive with a 95-percent paired interval above zero at every one of the four ranks, in both disciplines. The rank contrast is H_{90} minus H_{25} for east, north and radius over each saved fit range. The directional contrast is north minus east at each rank over the full range. Both retain pairing inside the flight bootstrap. The zero line marks the stated equality hypothesis, without a multiple-comparison calibration.

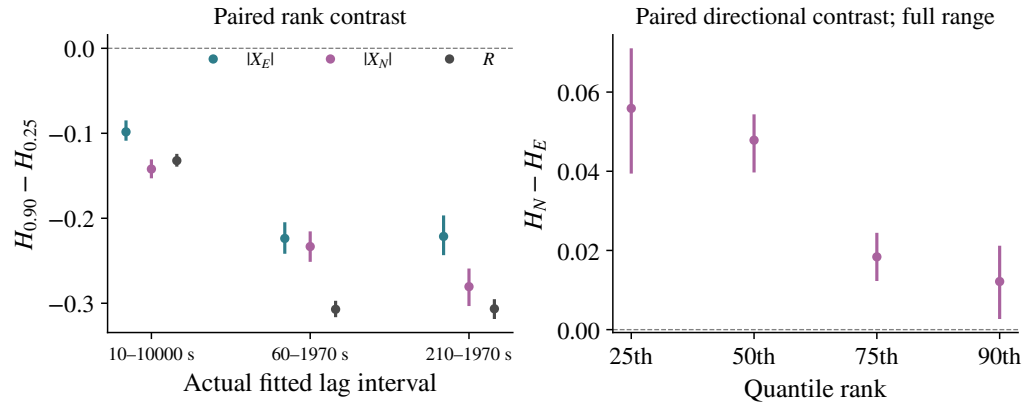
Hang gliders: component and radial exponent estimates

Markers are full-range fitted exponents; vertical segments are the exact saved 95-percent percentile endpoints from 400 whole-flight bootstrap draws. They are not simultaneous intervals and do not include sensor, site-day or fitting-range uncertainty.



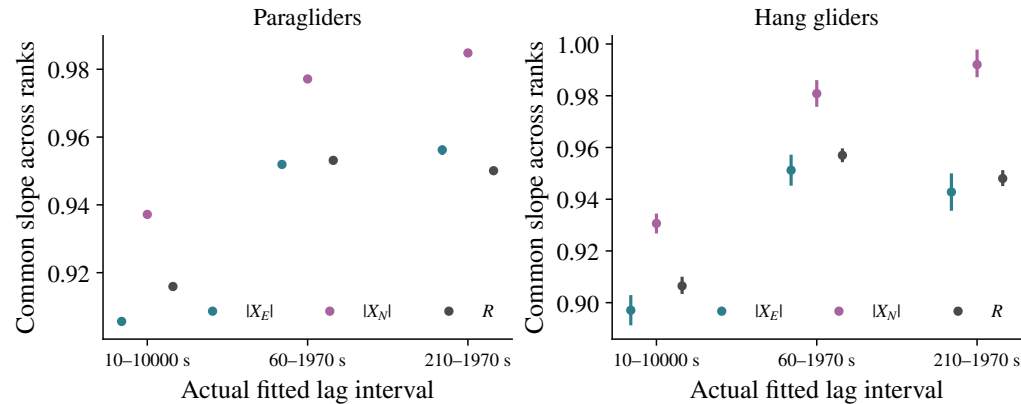
Which differences need a paired contrast instead of comparing overlapping intervals?

Hang gliders: paired rank and directional contrasts



The full-range north-minus-east contrast is positive with a 95-percent paired interval above zero at every one of the four ranks, in both disciplines. The rank contrast is H_{90} minus H_{25} for east, north and radius over each saved fit range. The directional contrast is north minus east at each rank over the full range. Both retain pairing inside the flight bootstrap. The zero line marks the stated equality hypothesis, without a multiple-comparison calibration.

Which differences persist under a change of fitting interval?



Which range has an independent physical justification?

A common fitted scale depends on the fitted interval

The component-specific common slopes average the four ranks with their own intercepts. The plot uses saved fits and intervals only, not new fitting. Agreement of a common summary does not establish equal individual rank slopes.

$$\hat{f}_j = \frac{p_j}{b_{j+1} - b_j}.$$

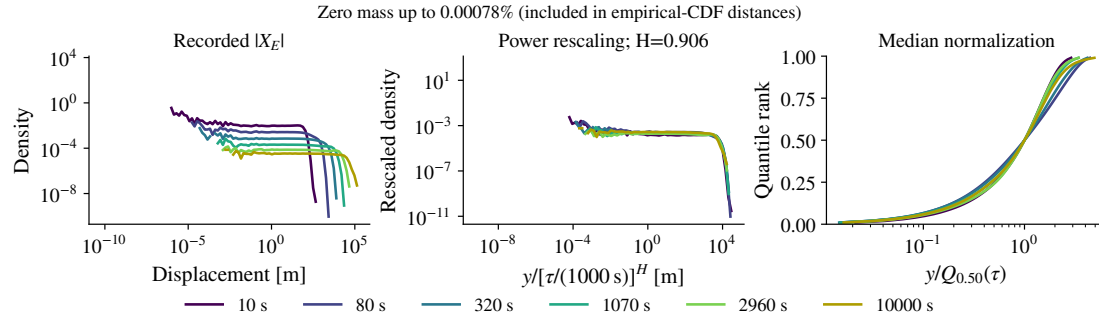
A very narrow histogram bin can have a tall density and still contain little probability. Across the displayed marginal laws, the mass below 1 mm, including zero and the straddling bin, is below 0.003%.

That numerical scale is not a claim of millimetre GNSS resolution. The following plots retain positive support; empirical-CDF comparisons also retain zero mass.

Do the central ranks and the probability-bearing tails tell the same scaling story?

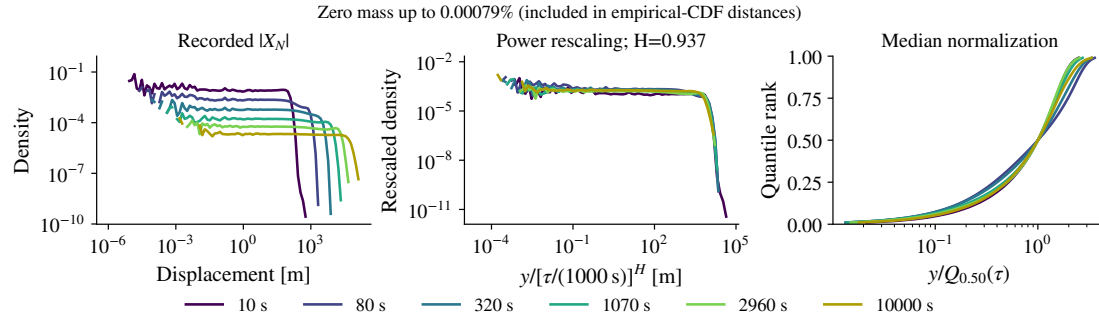
A large density near zero need not carry much probability

The stated bound comes from the complete saved histogram masses, not from reading a logarithmic density height. Tiny numerical displacements can reflect recorded coordinates and interpolation. The median-normalized displays show the saved first-to-ninety-ninth percentiles; they do not establish tail collapse beyond those ranks.

Paragliders: marginal collapse for east

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.

Does power or median normalization leave a systematic shape change?

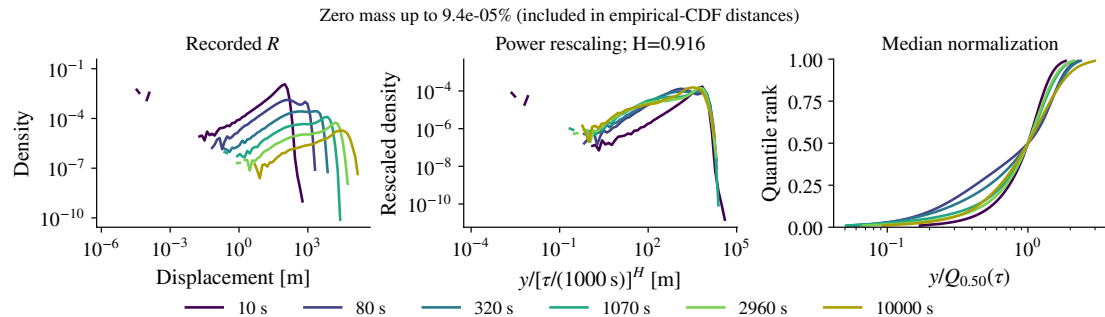
Paragliders: marginal collapse for north

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.

Does power or median normalization leave a systematic shape change?

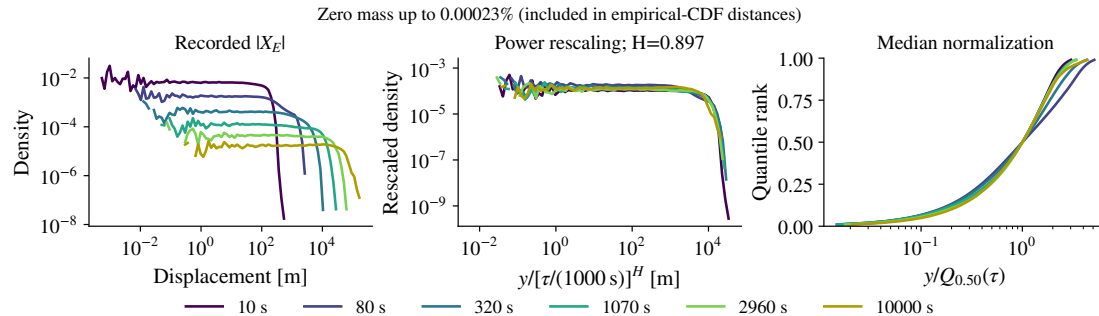
Paragliders: marginal collapse for radius

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



Does power or median normalization leave a systematic shape change?

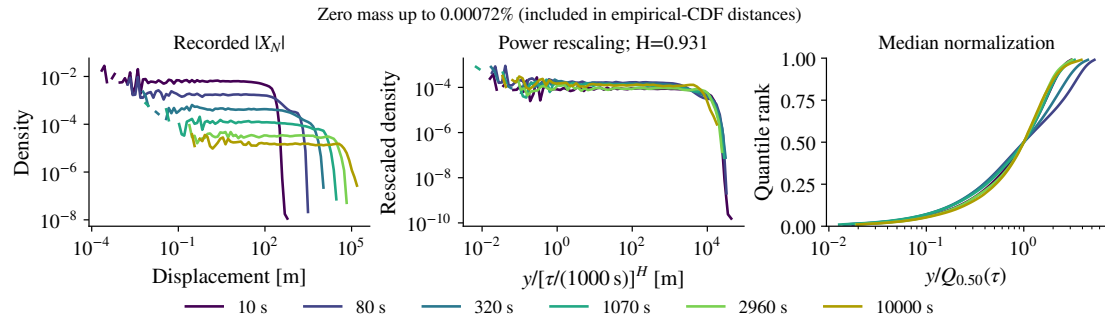
Hang gliders: marginal collapse for east



The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.

Does power or median normalization leave a systematic shape change?

Hang gliders: marginal collapse for north

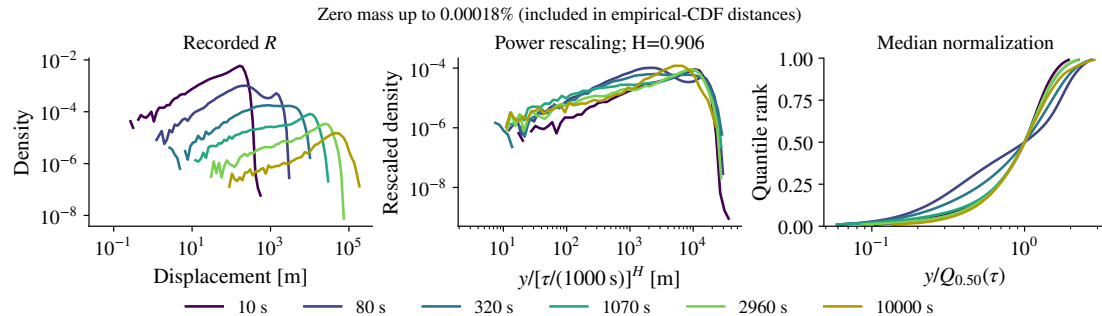


The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.

Does power or median normalization leave a systematic shape change?

Hang gliders: marginal collapse for radius

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



Does power or median normalization leave a systematic shape change?

$$\mathbf{Z}_\tau = \frac{(X_E, X_N)}{s_0 \left(\frac{\tau}{1000\text{s}} \right)^{H_*}}.$$

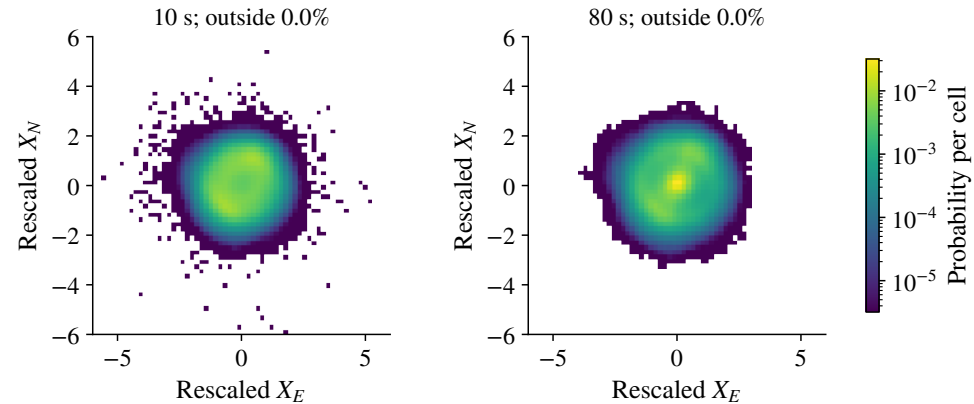
H_* is the common fit across all component/rank slopes. The radial reference median near 1000 s is adjusted to 1000 s by the same power.

There is no lag-specific centring, rotation or whitening. The six lags share the same cells and equal total weight per flight.

Which directional and dependence changes remain visible when the normalization cannot remove them separately?

The joint comparison keeps geographic axes and one scalar scale

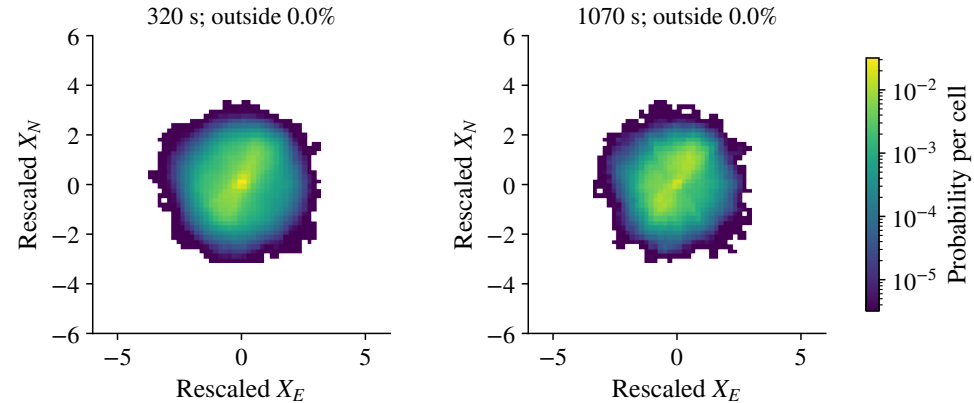
The common geographic axes are deliberate. Independent rotations or covariance normalization at each lag could erase part of the effect being studied. This is a test of one stated scale family of ground displacements, not a recovery of air-relative motion.



Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (1/3)

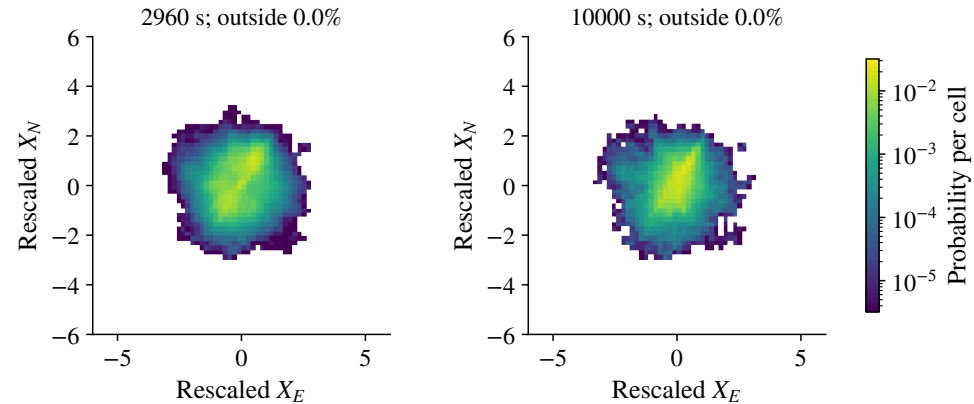
Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.



Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (2/3)

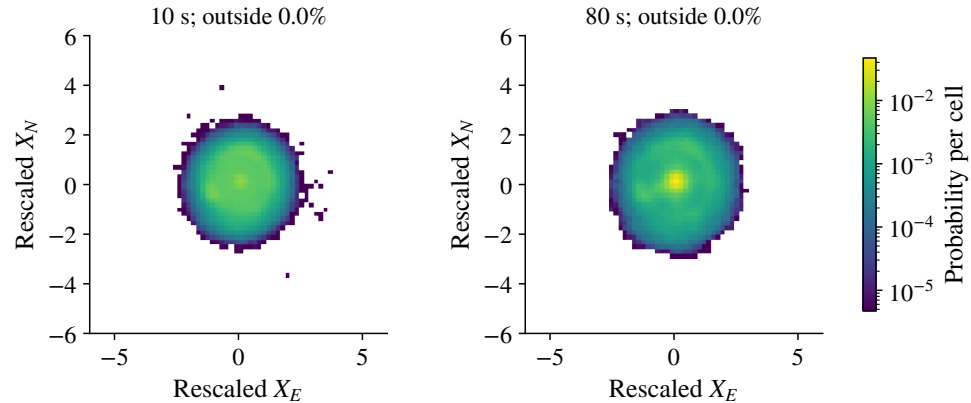
Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.



Which changes concern orientation, central concentration or mass outside the displayed square?

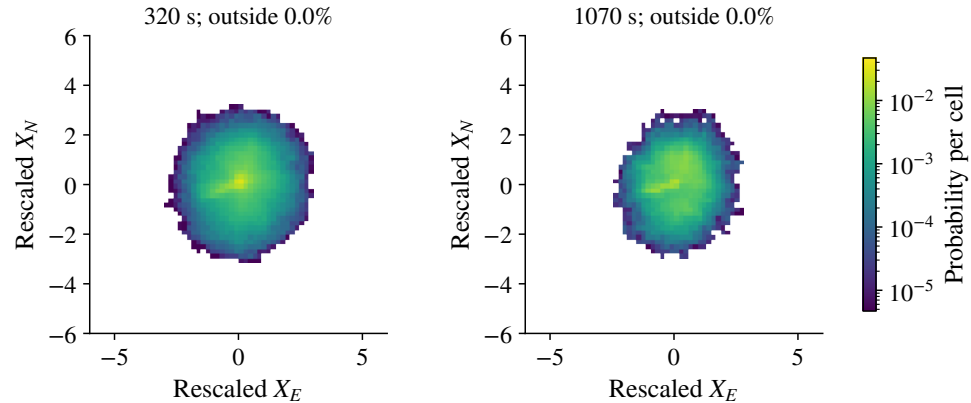
Paragliders: signed joint laws (3/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Hang gliders: signed joint laws (1/3)

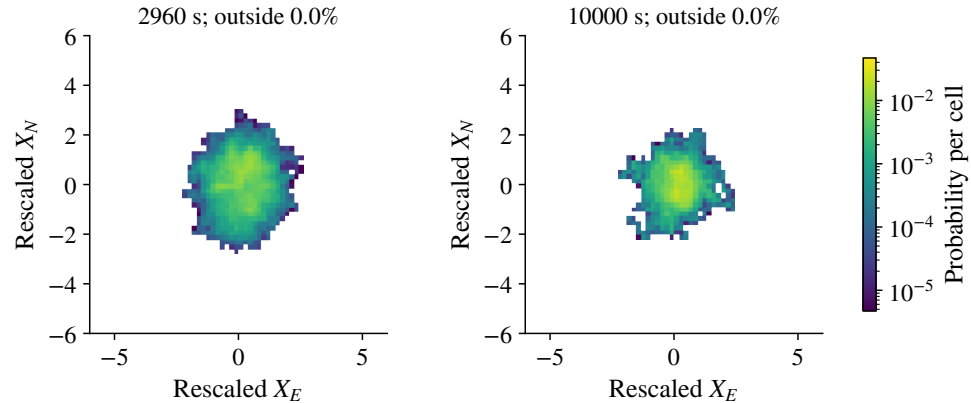
Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Hang gliders: signed joint laws (2/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Which changes concern orientation, central concentration or mass outside the displayed square?

Hang gliders: signed joint laws (3/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Which changes concern orientation, central concentration or mass outside the displayed square?

	Paragliders	Hang gliders
Common scalar exponent H_*	0.9196	0.9114
Binned TV: 10 s versus 10,000 s	0.370	0.403
Largest TV among the six lags	0.370	0.441

Later rescaled laws concentrate more centrally, while their means also change. Overflow mass is zero to numerical precision in the displayed square.

Mean motion, centred spread, orientation and dependence can all contribute. The plot alone does not separate their physical causes.

Would separating mean displacement from centred shape clarify which assumption fails?

One scalar rescaling leaves changes in the signed joint law

The comparison uses fixed flights, common origins, equal flight weights, signed geographic components and the same bins. These distances describe the recorded finite distributions; they have no calibrated rejection threshold. The first-to-last pair is not the maximum pair for hang gliders.

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|.$$

The grid has 64 interior bins per axis and overflow cells. The distance resolves changes at that bin scale; it is not a calibrated rejection test.

Even agreement of every one-lag vector distribution would leave the joint law at several times, and hence full process self-similarity, untested.

Which multi-time or phase-conditioned comparison would most help choose between candidate processes?

One-lag vector laws leave multi-time questions open

Finite empirical distributions can differ under a common population law. Calibration must respect paired flights and dependent origins. Energy distance is an alternative multivariate diagnostic under finite first moments, but choosing another distance does not automatically supply the right uncertainty model.

Two groupings

Alps, Pyrenees and Channel Coast use the defined geographic boxes.

Initial GNSS altitude h_0 defines:

Plains	$h_0 < 300$ m
Hills	$300 \leq h_0 < 800$ m
Low mountains	$800 \leq h_0 < 1500$ m
High mountains	$h_0 \geq 1500$ m

One flight-level estimator

$$m_f(\tau) = \frac{\sum_s n_{fs} m_{fs}}{\sum_s n_{fs}},$$

$$M_g(\tau) = \frac{1}{N_g} \sum_{f \in g} m_f(\tau).$$

Pool supported origins inside each flight, then weight contributing flights equally.

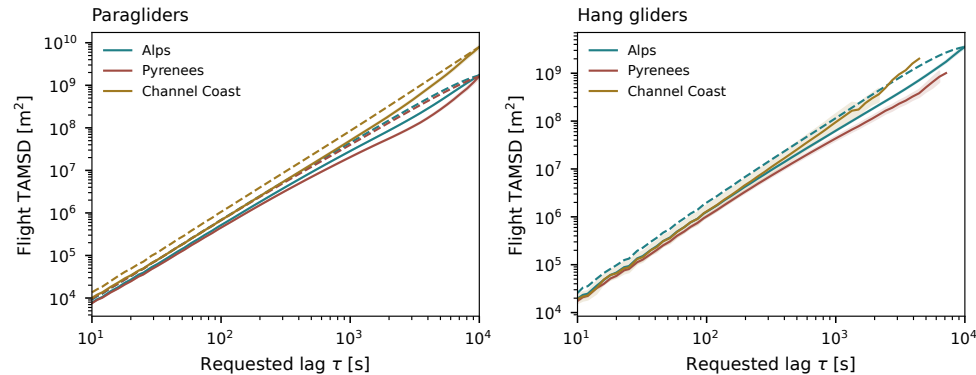
Compare available segments with a fixed set supporting the largest lag.

How much of a group contrast concerns transport, and how much concerns which flights remain?

Native-grid TA-MSD; 155,085 para / 6,060 hang flights. Altitude groups include the full eligible archive.

Compare transport magnitude before its directional structure

The altitude variable is the cleaned GNSS coordinate origin, with no pressure fallback. The chapter-two exploratory map used the raw first altitude with a fallback, so the two grouping sources must be distinguished. The familiar names do not measure relief, terrain encountered along a route or instantaneous flight altitude. Native intervals are at most ten seconds, each segment has at least eight fixes, and stored lag support ends at half the segment's grid length. Requested lags are rounded to native steps. No displacement crosses a gap. The fixed control retains identical segments and flights over all lags, but available time origins and their segment weights still vary.

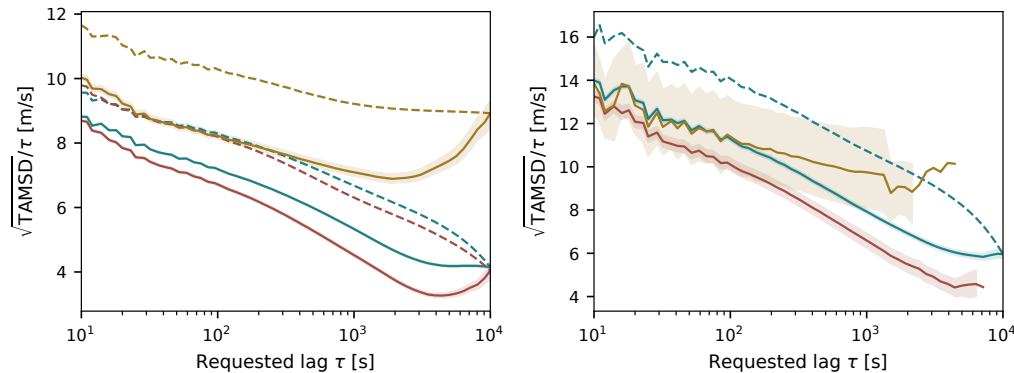
Regional TA-MSD: coastal flights travel farther at intermediate lags

Does the coastal ordering persist after restricting all lags to the same long segments?

Solid: available segments. Dashed: fixed long segments. Shading: pointwise 95% site-day bootstrap intervals for solid curves.

At the requested 1049-second lag, paraglider RMS displacements are 5.54 km in the Alps, 4.69 km in the Pyrenees and 7.42 km on the coast. The fixed-segment values are 6.97, 6.58 and 9.65 km. The coastal ordering remains, whereas the mountain contrast narrows. These are RMS endpoint displacements, not path lengths. Hang gliders have the same available-population ordering, but the coastal sample contains only 26 flights. Lags are the native TAMSD grid; 1049 seconds must not be labeled an exact 1000-second PCA lag.

Fixing the population changes the apparent long-lag growth

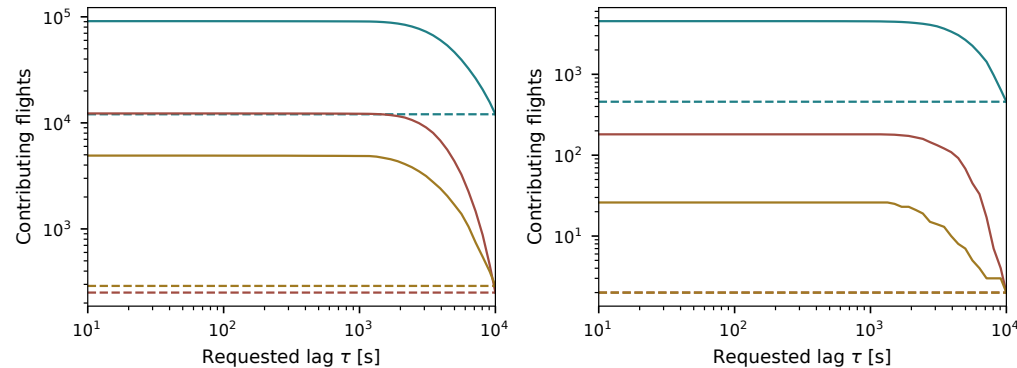


Would the upward turn remain if short flights could contribute observed long-lag displacements?

The ordinate is RMS interval-mean ground speed, up to native-grid rounding. A horizontal curve corresponds to quadratic TAMSD growth.

Fixing the population changes the apparent long-lag growth

The available coastal paraglider TAMSD has a descriptive long-range slope of 2.20; the fixed-segment curve has 1.98. The superquadratic appearance therefore does not persist in this fixed-population fitted comparison. It is not evidence for an intrinsic superballistic process. The Alpine and Pyrenean fixed curves continue to fall in this normalization. Their higher intermediate-lag amplitude also shows that long-flight selection changes the observed population before short flights lose support. These are descriptive interval fits, not Hurst estimates or calibrated significance tests.

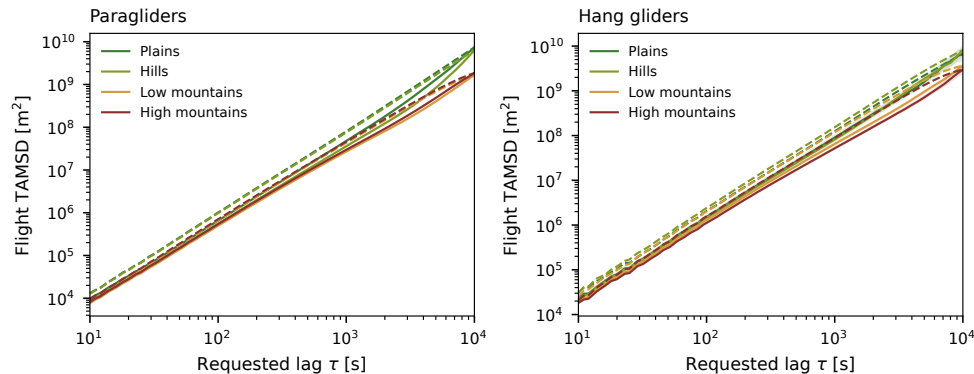


Which regional conclusions have enough long-flight support in both disciplines?

Fixed paraglider flights: 12,053, 251, 290.

Regional support limits what the fixed control can establish

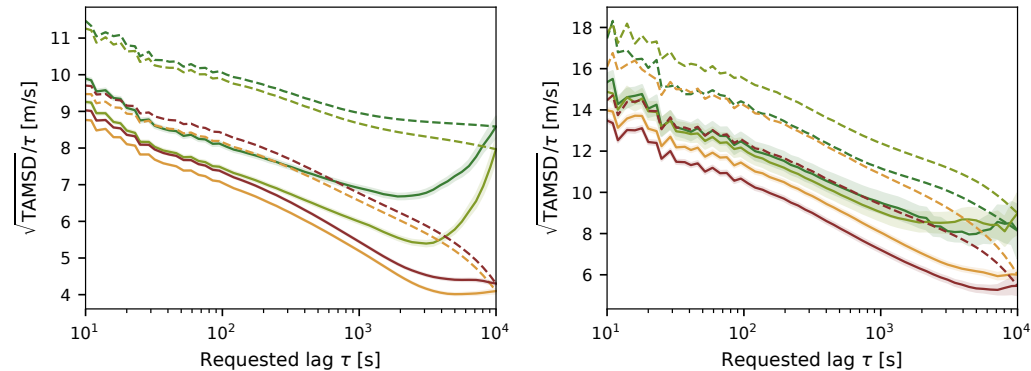
The solid curves show changing support and the dashed curves fixed membership. The fixed Alpine hang-glider control contains 458 flights, but the Pyrenean and coastal controls contain only two each. Those two-flight means are saved but omitted from displayed population curves. Eight flights is a display minimum, not a precision guarantee. Intervals require at least twenty site-day clusters; overlapping origins are never independent bootstrap units.

Initial-altitude groups combine transport and regional composition

Does an altitude-band contrast remain when region and contributing segments are held comparable?

Bands use the cleaned GNSS origin: Plains, Hills, Low mountains, High mountains. All eligible regions enter these curves.

At 1049 seconds the available paraglider RMS values are 7.23, 6.25, 5.40 and 5.66 km. The altitude ordering is not monotonic: the highest band exceeds the lower mountain band. For hang gliders the values are 9.91, 9.42, 8.38 and 7.49 km, but those groups differ in geographic and flight composition. Neither ordering is a within-pilot effect of altitude. The next panels expose the effect of fixing long segments and the lack of comparable regional-altitude populations.

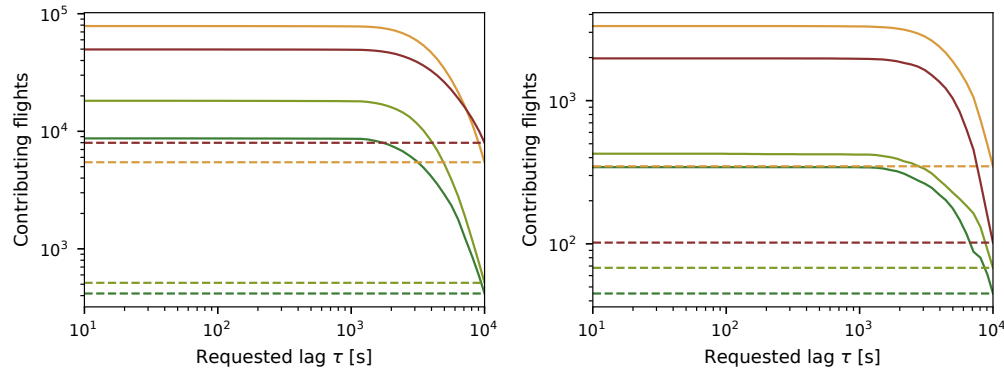


Which parts of the ordering survive the fixed long-segment comparison?

Solid: available segments; dashed: identical long segments at every lag. Shading is pointwise, not simultaneous.

The late rise in Plains and Hills is sensitive to population selection

Paraglider long-range slopes change from 2.19 to 1.97 in Plains and from 2.21 to 1.93 in Hills after fixing the contributing segments. Their RMS amplitudes remain above the mountain-labelled groups near the reference lag, but their separation becomes smaller. For hang gliders, the Plains/Hills ordering reverses in the long-segment population. These comparisons show selection sensitivity; they do not isolate an altitude or terrain mechanism.



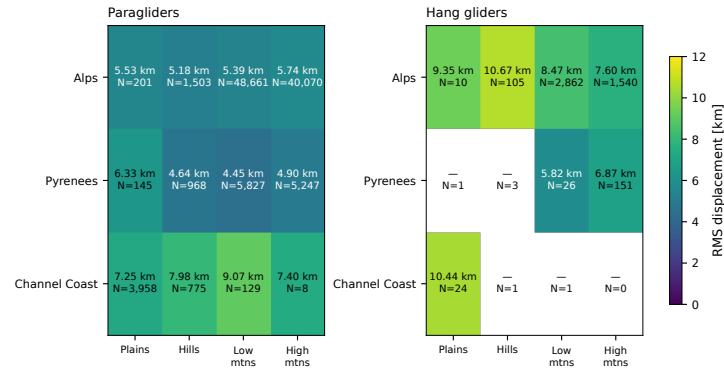
How much does the balance between altitude groups change as the lag increases?

Counts concern supported flights. The fixed-segment control retains membership; its origins still depend on lag.

The altitude-band comparison also needs its flight counts

Most eligible flights fall in the two mountain-labelled altitude bands, and long-flight retention differs greatly between bands. The paraglider fixed counts are 417, 514, 5452 and 7982; hang-glider counts are 45, 68, 348 and 102. The fixed control is conditional on long continuous segments, approximately twenty thousand seconds, with exact admission determined by the native-grid half-length rule. This explains small count differences from exact common-grid controls.

Region and origin altitude are strongly associated in this sample



The coastal sample mainly occupies Plains, while the mountain regions mostly occupy the two higher bands. Within Alpine paragliders, altitude-band amplitudes are much closer than in the pooled altitude comparison. The few coastal records with high initial altitude do not prove high terrain; records can begin airborne. Values below eight flights are omitted, with counts retained. Later, the coastal V2 change is smaller despite larger TAMSD: that is compatible with more persistent net advancement, but the two diagnostics use different lag grids and support, so it is a qualitative comparison. Independent wind and route information are needed for attribution. We now examine directionality, which a scalar TAMSD cannot identify.

Can a pooled altitude contrast be interpreted as terrain when the same strata barely overlap?

RMS displacement at the requested 1049-s lag; available segments. Descriptive cells, with no calibrated intervals.

Diagnostic	Clock	Weighting	Centring
Launch-time ratio	time since launch t	paired flights at each t	raw moments
Regional PCA	increment lag τ	eligible origins pooled	mean removed
Fixed joint law	increment lag τ	equal total flight weight	mean retained

These measurements differ in method as well as population. Compare these choices before attributing a difference to a physical mechanism.

Which apparent disagreement remains after comparing like populations and observables?

Regional plots use different clocks, weights and centring

The PCA draws on the full eligible regional archive, not only the fixed long-flight cohort. Longer supported records supply more increment origins to that pooled covariance. The launch-time ratios use a different clock and a changing population. Make these distinctions before linking the three figures to wind or terrain.

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1 / \lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

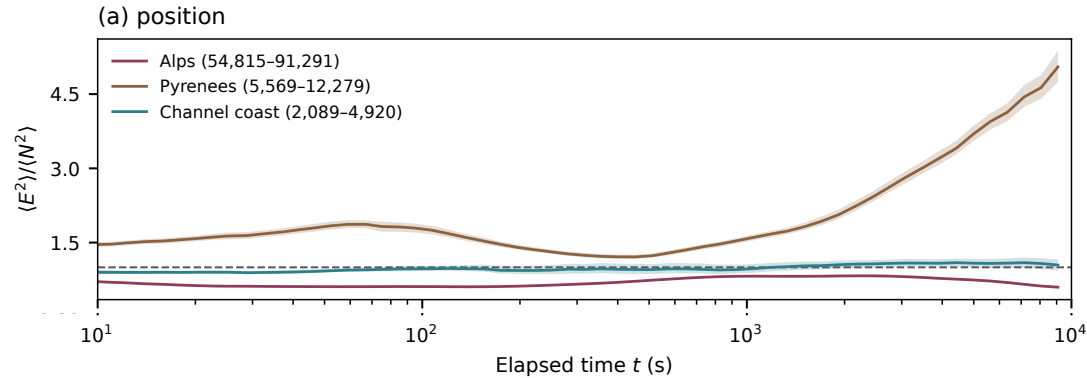
Are we observing mean drift, centred directional variability, or a mixture of individually oriented routes?

Equal east and north moments do not establish isotropy

Full isotropy requires rotation invariance of the law. An isotropic covariance alone would still be only a second-order condition. Raw moments additionally contain the outer product of the mean with itself. An intentional downwind route can be successful despite being directional, so this ratio is not a direct performance score.

Regional position moments differ along east and north

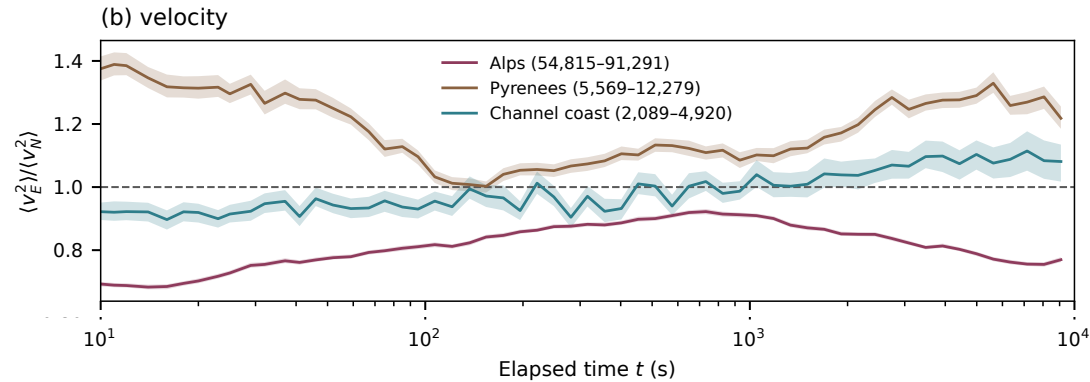
The clock is time since estimated launch, not an increment lag. Position ratios approach five in the Pyrenees. Raw second moments include both covariance and the squared mean. Splits prevent cross-gap stencils but recorded endpoint offsets can still affect these launch-referenced coordinates.



How much imbalance reflects mean drift, route orientation or changing support?

Regional velocity moments show a related directional contrast

The regional figure pools paired east/north observations at each elapsed time and uses site-day pointwise bands. Terrain, weather, routes and the disappearing short-flight population can all contribute. Coordinate-moment balance is weaker than full isotropy.



Which position and velocity contrasts persist under matched conditions?

The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

Would a paired site–day comparison change the group difference and its uncertainty?

Regional uncertainty bands do not match the groups

The bands are not 95-percent intervals, simultaneous bands or direct intervals for the difference between groups. Their overlap is not itself a calibrated significance test. The many displayed time points are correlated and must not be counted as independent confirmations of an ordering.

A rotation preserves distance

$$\Sigma = U\Lambda U^\top, \quad |U^\top \mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

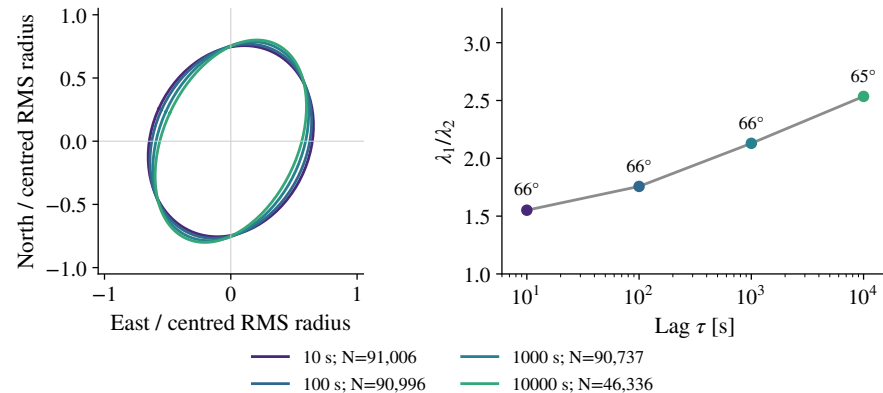
Different marginal exponents, or changing joint dependence, can prevent radial scaling.

Choose matched environments and an independent wind estimate before defining an “environment-free” model target.

The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

PCA describes preferred axes; wind needs independent information

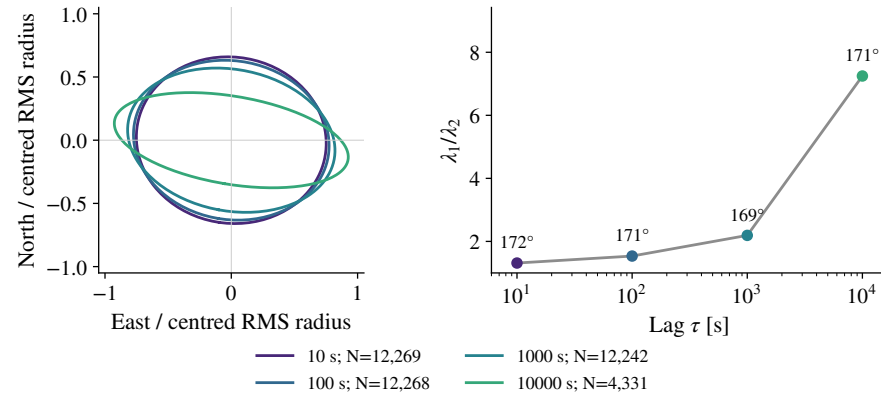
Even independent Gaussian components can yield a changing radial exponent if their component exponents differ. If both component exponents agree, changing dependence can still prevent joint scaling. Conversely, a joint vector scaling law always implies radial scaling, whether the vector is isotropic or not. Rotating each flight randomly would balance aggregate directions without changing radial displacements; it cannot recover still-air motion. Independent wind information or carefully justified phase-based wind estimates should be discussed before choosing an environmental correction.



Does the axis remain stable as lag and contributing population change?

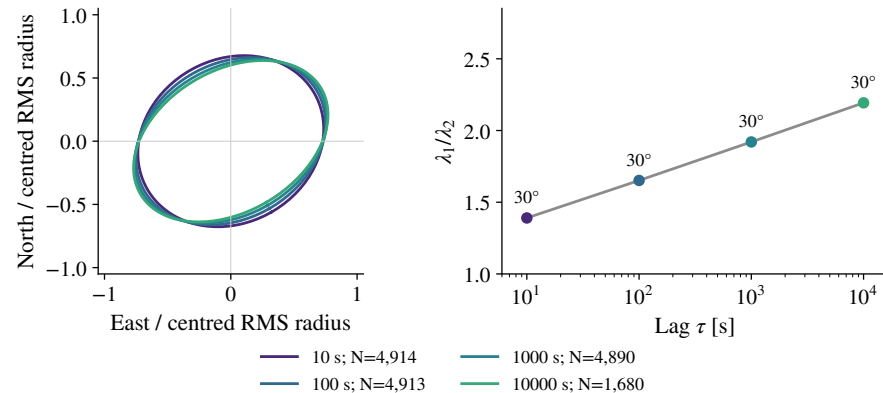
Alps: centred principal axes across four lags

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

Pyrenees: centred principal axes across four lags

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

How much of the stronger 10,000-s elongation persists when flight identities are held fixed?



Does the axis remain stable as lag and contributing population change?

Channel Coast: centred principal axes across four lags

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

NOAA ETOPO 2022

Same Alpine and Pyrenean take-off boxes.

One-arc-minute elevation sampling; retain the footprint above 1000 m.

Project cell centres to regional east/north coordinates. The area-weighted spatial covariance gives the main axis.

Definition sensitivity

Thresholds: 500, 1000, 1500 m.

Alps: 42.1–47.9°.

Pyrenees: 169.7–175.4°.

These spans are not confidence intervals; the axis describes an area, not a traced crest.

Would local axes along the actual tracks change the regional comparison?

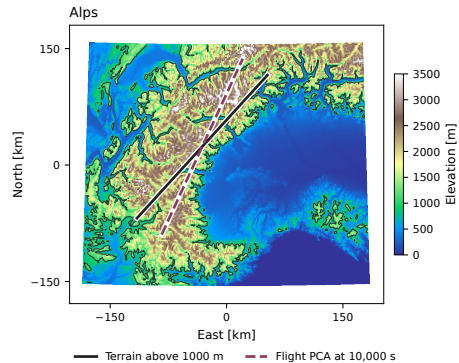
NOAA NCEI, ETOPO 2022. Both elevation subsets, requests, code and hashes are archived. Angles are counterclockwise from east, modulo 180°.

A terrain direction measured independently of the flights

Every fourth cell of the 15-arc-second grid gives one-arc-minute spacing. We select elevations at or above 1000 metres without fitting the threshold to the flight PCA. WGS84 tangent coordinates use the box centre and zero projection height. Weights are cosine latitude to approximate cell area; elevation above the threshold adds no weight. Threshold sensitivity is descriptive. The box boundaries, curved chains and plateaux affect the direction. Flight boxes identify take-offs, while increments can leave them and use flight-centred tangent frames. This is an indicative regional comparison.

Alps: shared sector, with an approximately 18-degree offset

Both orientations occupy the northeast-southwest sector, but the flight covariance is more northward. A simple claim that flights run parallel to one Alpine axis misses the approximately eighteen-degree difference. The terrain angle spans 42.1 to 47.9 degrees over the three thresholds, so this sensitivity does not eliminate the offset. Wind and long-flight selection remain possible contributors; the figure neither isolates a terrain effect nor tests pilot skill.



Terrain: 47.6° .

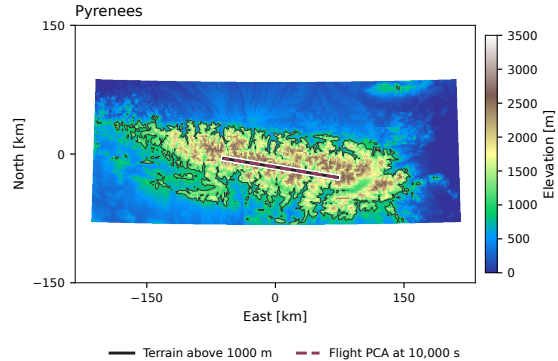
Flight PCA: 65.2° .

Separation: 17.6° .

46,336 flights support 10,000 s.

How much of the offset reflects Alpine curvature, box truncation or route selection?

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.



Terrain: 170.5° .

Flight PCA: 170.5° .

Separation: less than 1° .

4,331 flights support 10,000 s.

Pyrenees: the long-lag spread follows the highland axis

The near-parallel axes support a geographical association between displacement orientation and the range. They do not establish the causal contribution of terrain. Winds can align with the same range and pilots can select routes along it. The sensitivity span is not a sampling confidence interval. The comparison is made within the exact Pyrenean take-off box, but flight increments can extend beyond it.

Does the alignment persist for fixed flights and for terrain sampled along their routes?

ETOPO footprint above 1000 m; threshold sensitivity: $169.7\text{--}175.4^\circ$. Decimal agreement does not imply this physical precision.

Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Mean: 1,006 m

Equal flight weights: 980 m.

Middle half of window means:
854 m–1,160 m.Approximate altitude above sea level;
recorder datums are not harmonised.

Is one representative height sufficient, or should wind be matched to the altitude and location of each flight window?

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

Channel Coast: the reference height follows the PCA windows

The mean is 1006.262 metres across 2013 exact PCA windows from 1680 flights, with the same equal-window weighting as the covariance. Each window altitude is a trapezoidal time average of cleaned GNSS z interpolated onto the existing 10-second grid. No gap is bridged. Equal weight per contributing flight gives 980.260 metres; the window-mean quartiles are 853.932 and 1159.549 metres. Short flights do not define this target because they cannot contribute at 10000 seconds. The raw GNSS vertical datum is not harmonised, so MSL is an approximation; these numbers do not imply metre-level physical accuracy. The quartile range describes heterogeneity, not uncertainty in the mean. Matching the target height does not yet match flight dates, instantaneous positions or local atmospheric conditions.

2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

Main comparison: all year, all hours vs. all year, 09–17 UTC.

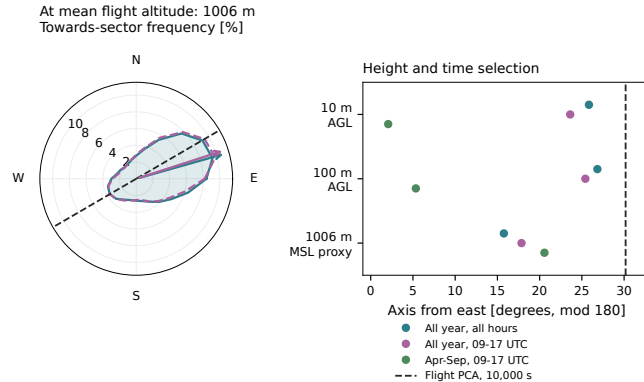
April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

How much of the surface-to-flight-height contrast reflects vertical shear, season and the flight conditions represented?

Copernicus/ECMWF ERA5 pressure levels, pinned Earthmover archive. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

ERA5 wind is interpolated to the measured flight altitude

Use the same nine grid cells as the surface analysis: latitudes 48.75, 49.75, 50.75 degrees and longitudes minus 1.25, zero, 1.25 degrees. Geopotential height is Phi divided by 9.80665; geometric height uses spherical Earth radius 6371000 metres. A pressure level changes altitude from hour to hour. Linear interpolation acts on eastward and northward velocities, never direction angles. All 789048 cell-hour profiles bracket the main target above the model surface. Average components over time and with cosine-latitude cell weights, then reduce the towards-angle modulo 180 degrees for comparison with PCA. The annual all-hour selection has 87672 hours per cell; 09 through 17 UTC inclusive has 32877. The two warm-season selections have 43920 and 16470. UTC is fixed, without a daylight-saving correction. Source values, timestamps, coordinates, pressure levels, geopotential, surface geopotential and hashes are preserved losslessly for offline reuse.



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC. Why is the seasonal contrast larger near the surface?

Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

At flight height, both annual references remain offset from PCA

At the mean flight altitude the all-year all-hour axis is 15.8 degrees and the 09–17 UTC axis is 17.9 degrees, against 30.2 degrees for flight PCA. The corresponding separations are 14.4 and 12.3 degrees. At 10/100 metres AGL the annual axes lie closer to PCA, but their warm-daytime references rotate further away. At flight altitude the warm-daytime axis is 20.6 degrees, giving a smaller 9.6-degree separation. The altitude-matched comparison therefore has a less close annual alignment and a smaller time-selection sensitivity than the surface comparison. These are descriptive contrasts, not formal tests of different angles. The rose uses weighted 15-degree sectors, excludes speeds below 0.5 m/s only from frequencies, and retains all hours in vector means. Teal is all-year all hours; plum is all-year daytime. Green appears only in the right-hand seasonal check.

At 1,006 m	Wind axis	PCA gap	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
Apr–Sep, all hours	16.0°	14.2°	0.35
Apr–Sep, 09–17 UTC	20.6°	9.6°	0.37

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

Does the remaining offset reflect flight selection and route choices, or change when weather is matched to actual flights?

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$ measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

Height and time checks preserve a broad orientation resemblance

The annual all-hour and daytime resultant fractions are 0.43 and 0.44; warm-season values are 0.35 and 0.37. These do not give confidence in a direction or the fraction of observations near it. At the lower quartile height, 868 profiles at one cell have no above-ground pressure level below the target. For the height sensitivity only, exclude these 868 hours from all nine cells at every tested height, including the mean-height reference: 86804 common hours remain per cell. Apply the same seasonal and daytime masks afterward. On common support the annual quartile-height axes range from 14.3 to 17.2 degrees for all hours and from 16.4 to 19.3 degrees for daytime. Equal-flight mean altitude changes the matched reference by less than 0.3 degrees, and common-hour selection changes the main annual reference by less than 0.1 degrees. These checks retain the offset but do not calibrate receiver datums or establish significance.

Region	Conclusion supported by this comparison
Pyrenees	Close terrain–flight alignment; a geographical association worth testing along tracks.
Alps	Same broad sector, with an approximately 18-degree offset from the mean highland axis.
Channel Coast	Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance.

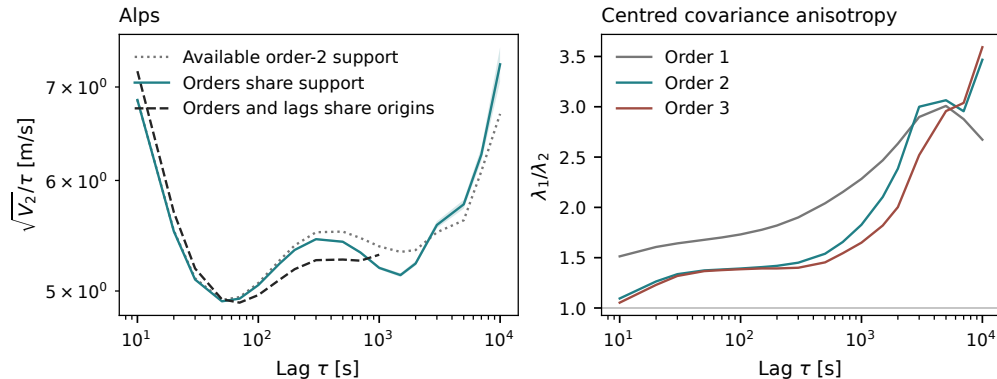
A common constant wind shifts the mean displacement by $\tau\mathbf{w}$; centring removes it. Variable winds and route responses can still shape covariance.

Match flight times, heights and routes to wind and local terrain before attributing the ellipse or equipment contrasts to a cause.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

What the environmental comparisons establish

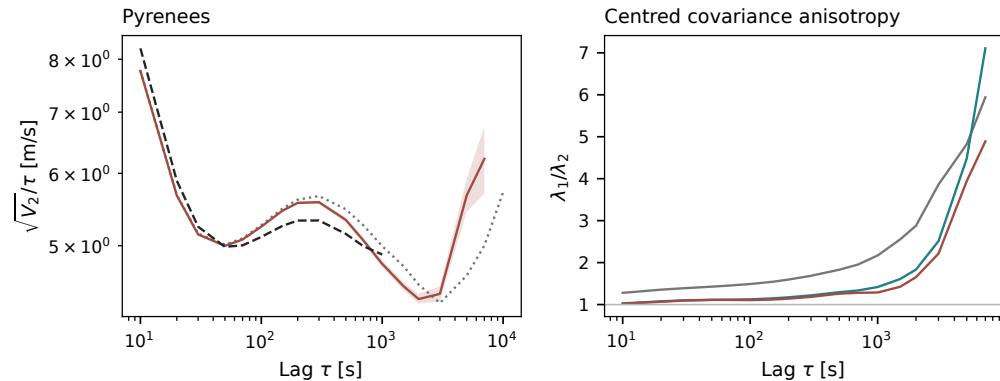
The PCA concerns centred displacement spread, whereas the wind reference concerns a mean velocity. Their orientation resemblance alone cannot show that mean wind causes covariance anisotropy. Constant additive drift cancels when displacement is centred at fixed lag. Variable drift between flights, time-varying forcing and route decisions can alter covariance; their contributions need matched observations. The Pyrenean agreement is the strongest geographical alignment here. The coastal comparison at mean flight altitude retains a broadly similar sector across the four time selections, with offsets of about 10 to 14 degrees. Surface winds exhibit a larger seasonal change, so their annual near alignment should not be transferred to flight altitude. Equipment class is not measured pilot expertise, and this comparison cannot establish resistance to wind.

Alps: velocity changes retain a directional structure

Left: available-order-two origins dotted, matched orders solid, and common origins through 1000 seconds dashed. Right: covariance anisotropy of orders one, two and three on identical origins. Unlike the preceding pooled-origin PCA, these curves give equal weight to flights. The full report also saves the pooled-origin sensitivity. Eighteen lags cover 10 to 10000 seconds. Matching all orders requires three times the lag; common origins use every ten-second origin supporting a 3000-second stencil. This changes the observations as well as flight membership.

Which part of the lag dependence persists with common origins and after changing difference order?

Equal flight weights. At 1000 s, matched orders retain 89,181 flights. Shading: pointwise 95% site-day intervals.

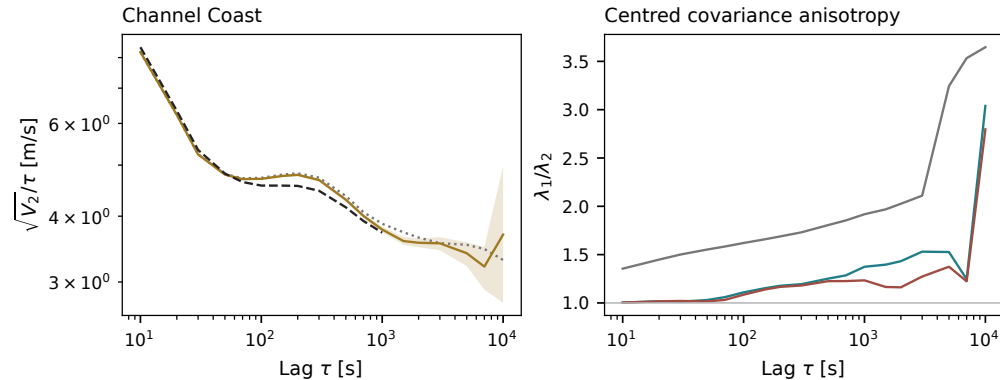


How much of the late rise survives a comparison with fixed flights and origins?

At 1000 s: 12,020 matched flights; at 10,000 s: only 3. Cells below eight flights are omitted from plots.

Pyrenees: the long-lag change coincides with falling support

The conventions are identical to the Alpine panel. At 1000 seconds the order-two eigenvalue ratio is 1.42, compared with 2.17 for ordinary increments. The sharp late changes need to be read together with the support figure. All numerical cells are retained in the report, including the final three-flight cell, but that cell is not plotted as a population result. The omitted 10000-second order-two ellipse is therefore intentional.

Channel Coast: smaller changes at 1000 s, with residual anisotropy

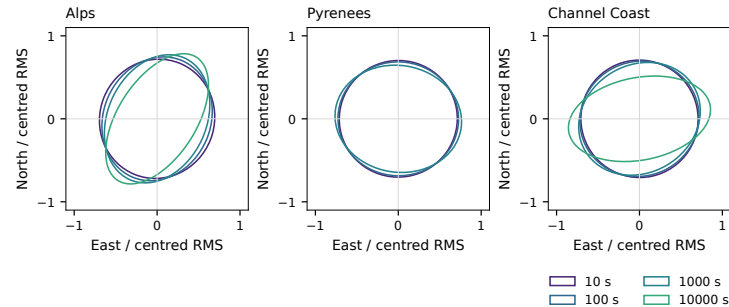
Could route geometry and varying forcing explain the contrast that survives constant-velocity cancellation?

At 1000 s: 4,679 matched flights, $D_2 = 3.77$ m/s and covariance ratio 1.37.

The coastal RMS change becomes lower than the mountain values over part of the lag range. It is higher than the Alpine value at ten seconds, so one amplitude ratio does not describe every scale. An added constant wind alone cannot explain a V_2 difference between otherwise identical populations. Variable wind, its effect on decisions, route geometry and selection remain possible. These boxes classify take-off locations, not exposure along the whole flight.

Second-difference ellipses retain orientation after drift cancellation

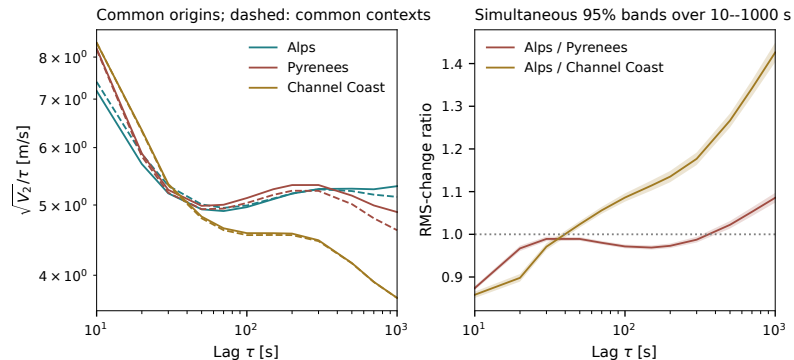
$$C_2 = \text{Cov}(A_2 \mathbf{r}), \quad V_2 = \text{tr } C_2 + |\langle A_2 \mathbf{r} \rangle|^2.$$



Do the resolved axes align with independent terrain and wind information?

Matched-order origins, equal flight weights; each ellipse is normalised by its own centred RMS.

The four colours match the exact PCA display lags. Ellipses describe centred spread and omit both amplitude and mean. Near-circular ellipses, especially the coastal ten-second one, do not resolve a useful direction. At 1000 seconds the order-two axes are approximately 65 degrees in the Alps, 171 degrees in the Pyrenees and 31 degrees on the coast, counterclockwise from east. Axes are modulo 180 degrees and do not supply signed wind directions. The preceding independent comparisons provide regional references; they do not measure terrain or weather exposure along individual tracks, and the coastal mean-height wind reference retains a 10–14-degree offset across the tested time selections.

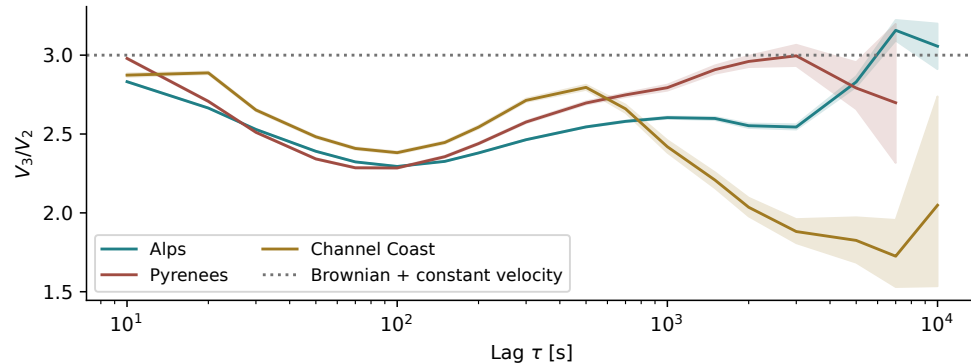


Which part survives matching actual weather, sites and pilot identity?

Common origins through 1000 s. Dashed: 23 shared context strata, descriptive. Shading: simultaneous 95% bands for the unstandardised ratios.

The regional contrast persists in a common-context control

The context control standardises task, retained-duration band, season period and equipment, with at least ten flights per region per retained stratum. Target stratum mass is proportional to the smallest regional count. It retains 65537 Alpine, 9933 Pyrenean and 4366 coastal flights. The 1000-second RMS changes are 5.13, 4.62 and 3.72 m/s. These dashed point estimates have no calibrated intervals. The shaded ratio intervals instead apply to the unstandardised common-origin population and use 500 site-day bootstrap replicates jointly over the supported lag grid. The Alpine-to-coastal ratio there is 1.426 with a simultaneous band from 1.403 to 1.449 at 1000 seconds. No equivalence margin was specified, so a non-significant difference would not establish equality.



Which scale interval could satisfy both the slope and amplitude predictions of one model?

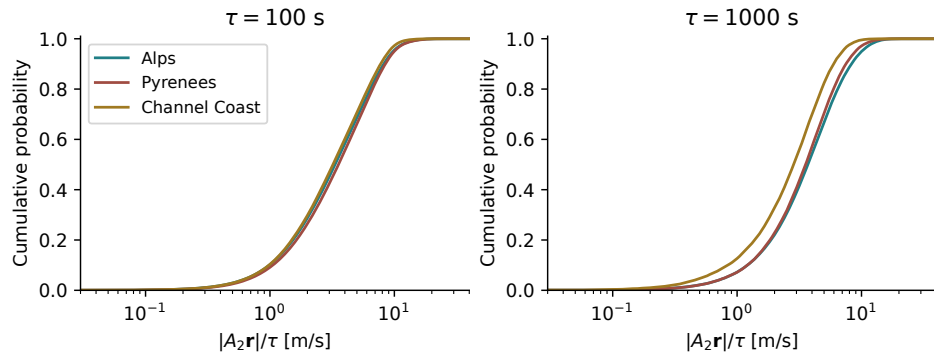
Matched origins and flight weights; pointwise 95% site-day intervals. Dotted: Brownian motion plus constant velocity.

The ratio between orders changes with the lag

V_3/V_2 is approximately 2.29 in both mountain regions at 100 seconds; at 1000 seconds it is 2.60 in the Alps, 2.79 in the Pyrenees and 2.42 on the coast. The lag dependence is itself diagnostic. V_3 is not an independent sample and does not automatically isolate still-air dynamics. Its longer support and larger sensitivity to position noise must accompany any interpretation. No Hurst estimate is validated by these curves.

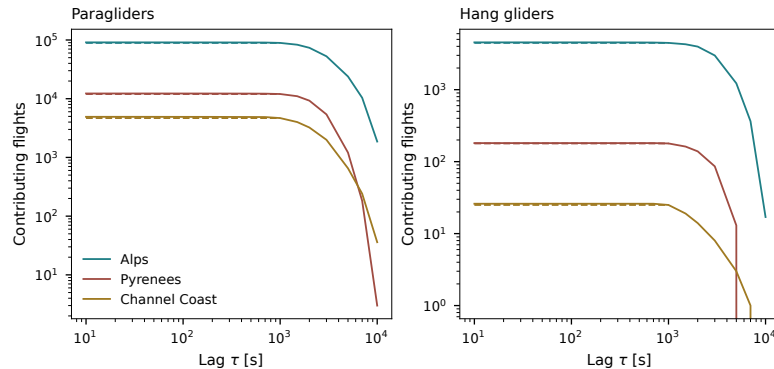
The distribution distinguishes frequent small changes from large events

At 1000 seconds the upper tenth of changes contributes about 43–47 percent of the squared-change signal in the Alps, 40–44 percent in the Pyrenees and 38–43 percent on the coast. Those ranges bound the unresolved histogram boundary bin; they are not sampling confidence intervals. The result measures concentration of this observable. It does not label turbulence or a particular flight phase. Phase conditioning and independent wind attribution have not been performed in this extension.



Where on the trajectories do the largest contributions occur, and which verified flight phases contain them?

Equal-flight CDFs on matched-order support. Full report: all four display lags, zero and overflow mass, and quantile bin bounds.

Order-three support limits the long-lag regional comparison

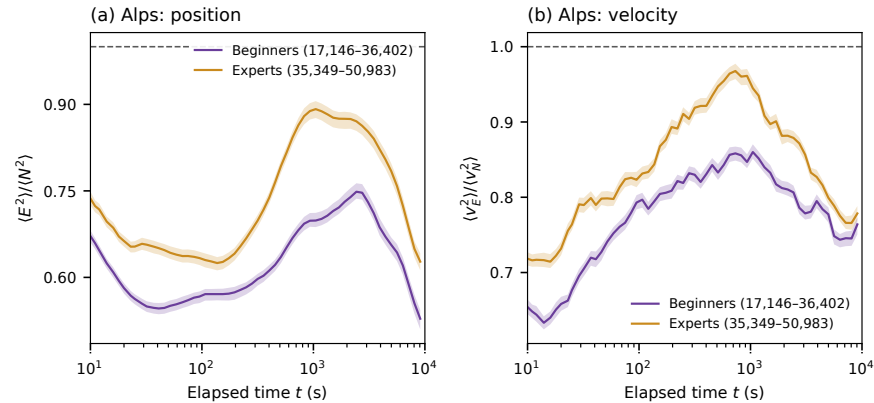
Which conclusions remain supported in both disciplines and across the same range of scales?

Solid: orders share 3 τ support. Dashed: origins common through 1000 s. All eligible regional flights contribute where supported.

The full regional measurement includes 108189 paraglider flights and 4747 hang-glider flights. At 10000 seconds, matched paraglider counts are 1865, 3 and 36 for Alps, Pyrenees and Channel Coast; hang-glider counts are 17, zero and zero. Coastal hang gliders number only 26 even at short lags. The report retains their estimates and suppresses intervals below twenty contributing site-day groups. No common three-region equipment/context strata survive for hang gliders, so that standardisation is not claimed for this discipline.

Alps: C/D/CCC has smaller coordinate-moment imbalance

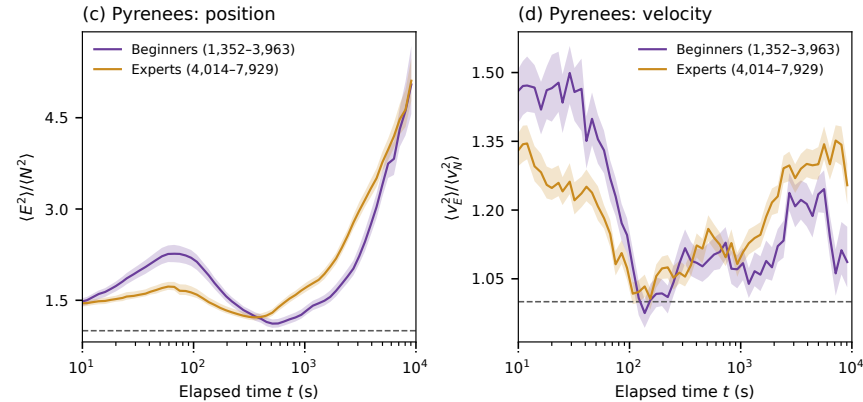
The fresh Alpine position and velocity ratios stay closer to unity for C/D/CCC over the supported grid. This is a descriptive ensemble observation. It does not establish individual isotropy, greater skill or a causal equipment effect. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Does the ordering persist after matching site, day, task and duration?

Pyrenees: the ordering of the two groups reverses

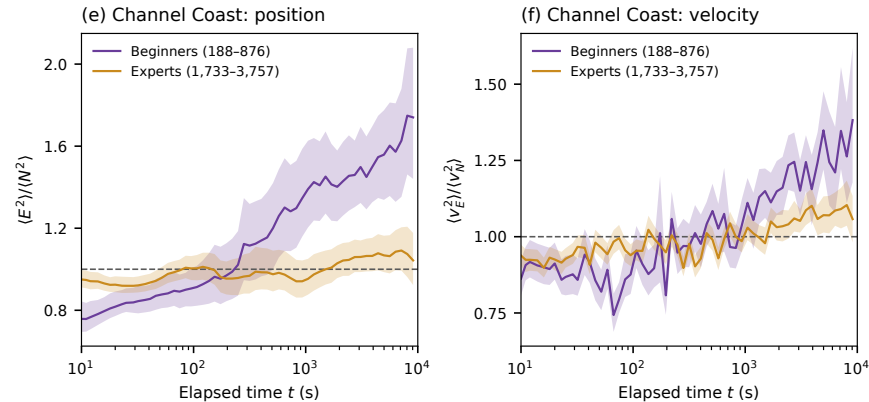
The Pyrenean curves do not support a uniform expert-proxy advantage. At late times both position ratios exceed five, while the velocity ordering also changes. Adjacent plotted times are correlated observations, not independent tests. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



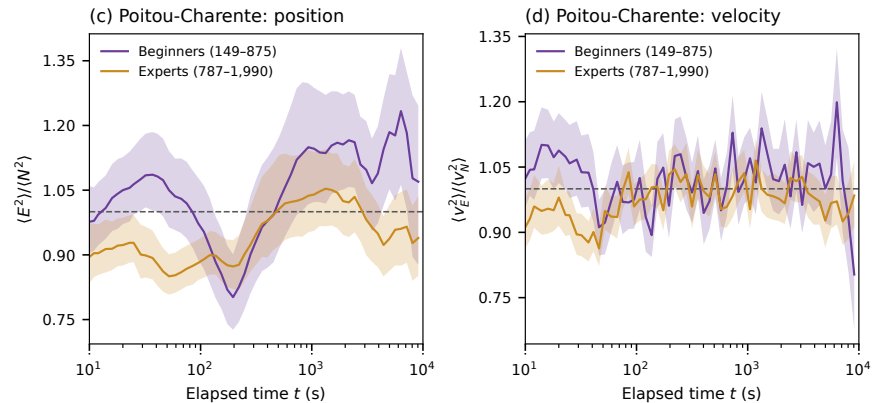
What changes between the early and later portions of the curves?

Channel Coast: the late position contrast is pronounced

Near 9086 seconds the position ratios are about 1.74 for A/B and 1.04 for C/D/CCC, supported by 188 and 1733 paired flights. These are raw ensemble second moments and the samples differ strongly. Coastal cliffs and valleys prevent a wind-only interpretation. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Can matching conditions separate directional route choice from wind response?

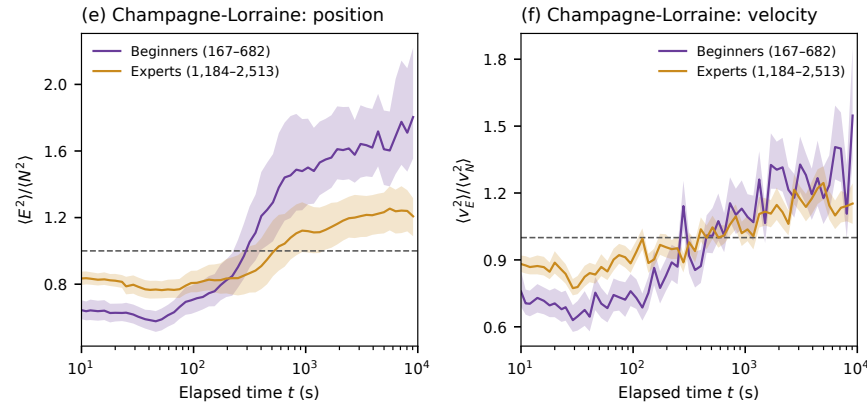
Poitou-Charente: the group curves cross

The crossing curves and broad overlapping bands do not support a general group ordering. These broad launch-location boxes are neither DEM-verified plains nor controls matched on weather. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

Does this low-relief launch box reproduce the coastal observation?

Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity

A later position contrast appears, but the C/D/CCC ratio also rises above one. Region, route mixture, weather and declining support must be separated before claiming the same physical cause. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Which parts of the coastal hypothesis transfer to this separate region?

Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

What observation would distinguish this hypothesis from different sites, tasks, weather or route mixtures?

The environmental interpretation remains a hypothesis

Preserve the proposed explanation as a useful hypothesis rather than silently removing it or presenting it as established. Similar curve shape does not by itself establish a common cause or an unavoidable terrain constraint. Wind and terrain can interact, so the coast/mountains split does not identify their separate effects.

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

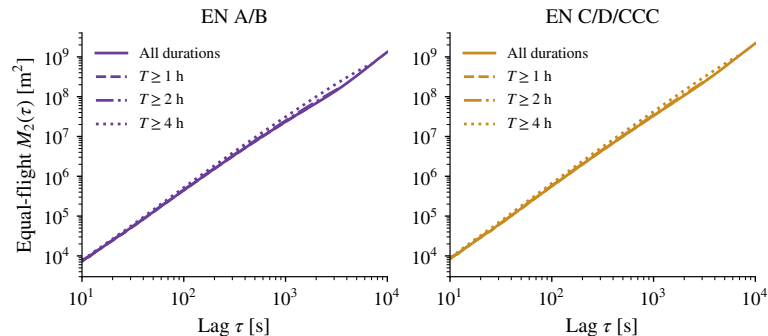
A first comparison would match overlapping site–day, duration and task support; then separate mean drift, centred variability and individual route orientation. Regional references show terrain alignment and a remaining coastal wind–PCA offset at flight altitude; local track-matched observations are still needed.

Which target and matched comparison should be the first priority?

Define the target before calling it resistance to wind

The site–day bootstrap adjusts one uncertainty structure; it does not perform matching or remove confounding. Where data permit, the same pilot on different equipment could help assess selection, but weather and task still require control. Differentiating velocity removes a constant wind term, but not its variation along the trajectory or the pilot's manoeuvres.

Long flights occur in both equipment groups



For $T \geq 4$ h: 7,415 A/B flights and 27,815 C/D/CCC flights.

Across 10–10,000 s, all-duration slopes are 1.728 and 1.788.

Equal-flight MSD; groups are nested. Threshold curves stop at $\tau = T_0/2$; at least 30 flights per lag. All counts and common-range slopes.

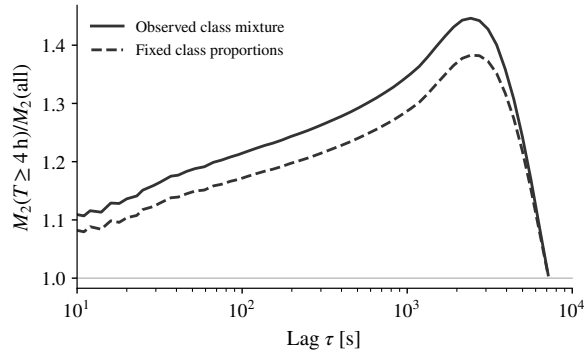
Long flights occur in both equipment groups

The hypothesis that only experts contribute to flights longer than four hours is contradicted by the counts. Experts nevertheless become more frequent in the long-duration groups. On the common 10–1695 second interval, expert slopes exceed beginner slopes in every duration cell; within both classes slopes also increase between all flights and the four-hour group. This shorter interval is needed to compare all four nested cohorts under the half-duration rule. It is distinct from the main three-decade fit. Equipment, pilot selection, region and weather remain possible causes.

Duration and equipment on the common 10–1695 s interval

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,292	1.755	0.022
Beginners	$T \geq 1$ h	51,124	1.754	0.022
Beginners	$T \geq 2$ h	31,768	1.767	0.020
Beginners	$T \geq 4$ h	7,415	1.806	0.017
Experts	All	94,283	1.803	0.017
Experts	$T \geq 1$ h	91,182	1.802	0.017
Experts	$T \geq 2$ h	69,087	1.811	0.016
Experts	$T \geq 4$ h	27,815	1.836	0.015

Counts are eligible flights, not independent windows; support falls with lag. RMS residual measures departures from a fitted line, not uncertainty. Threshold samples overlap.



$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau).$$

Fixed weights: $\pi_{A/B} = 0.365$,
 $\pi_{C/D/CCC} = 0.635$.

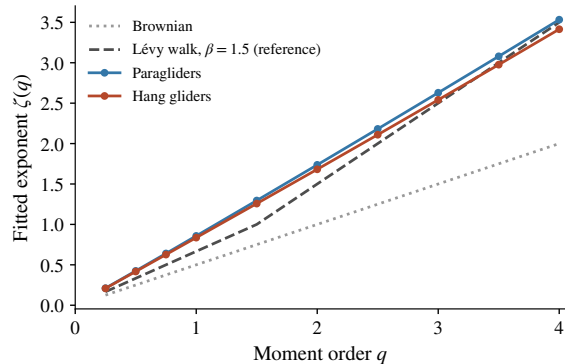
At $\tau = 1695$ s the ratio falls from 1.42 to 1.35. A within-class contrast remains.

Match site–day and task next. Convergence near 7200 s also reflects shorter flights leaving the reference.

Both curves use the same two known equipment groups. Equality at one lag does not establish independence from flight duration.

Fixing equipment proportions removes part of the duration contrast

The expert fraction among known classes rises from 63.5 percent overall to 78.9 percent in the four-hour cohort. Standardization uses the all-duration proportions at every lag for both numerator and denominator. It is a descriptive reweighting, not a causal adjustment. The remaining difference shows that the two-class proportions alone cannot account for the observed amplitude contrast. Near half the duration threshold, the reference loses shorter flights, so the curves are increasingly comparing similar populations.



$$M_q = \langle R^q \rangle \propto \tau^{\zeta(q)}.$$

For a constant-speed renewal walk with isotropic independent headings and $\psi(t) \sim t^{-1-\beta}$, $1 < \beta < 2$:

$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & q < \beta, \\ q + 1 - \beta, & q > \beta. \end{cases}$$

The empirical spectrum is nearly linear over $q = 0.25$ –4.

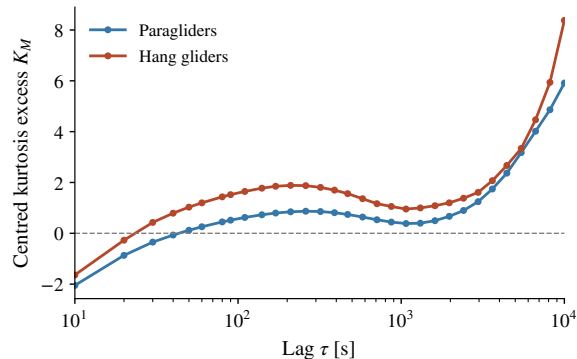
Can finite-record simulations resolve the predicted branch change under our sampling and selection?

Pooled windows. $\beta = 1.5$ is illustrative, not a fitted null; $q = \beta$ has a logarithmic correction. Zaburdaev, Denisov & Klafter (2015).

The moment spectrum challenges a specific Lévy-walk benchmark

The current pooled moment spectrum has $\zeta(2) = 1.737$ for paragliders and 1.682 for hang gliders. The corresponding $\zeta(q)/q$ ranges are imported in the report. The illustrative beta=1.5 line is not a fitted null. A finite-record comparison must reproduce sample lengths, cadence, selection and weighting, and assess whether the asymptotic branch change would be resolvable.

Non-Gaussian pooled increments can arise from Gaussian mixtures



$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}),$$

$$K_M = \langle |\mathbf{Z}|^4 \rangle - 8.$$

A nonsingular 2D Gaussian has $K_M = 0$, including anisotropic Gaussians.

The observed curve changes sign and rises at large lag, where support is weakest.

Test Gaussianity within matched environments or phases before rejecting conditional Gaussian dynamics.

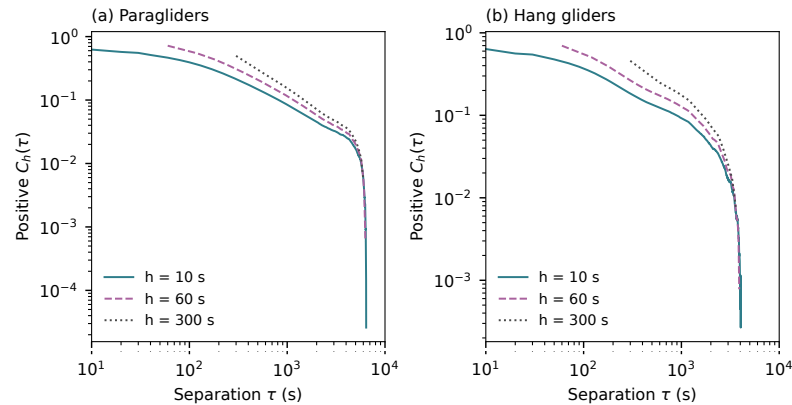
Centred, full-covariance Mardia excess; pooled windows. Positive excess does not identify a power-law tail. Mardia (1970).

Non-Gaussian pooled increments can arise from Gaussian mixtures

Explain the full covariance normalization: unequal east and north variances alone do not produce a nonzero population excess for a Gaussian. But mixing Gaussian variances does. If $\mathbf{X}$ equals $\sqrt{A}$ times a fixed Gaussian vector, K equals eight times $\text{Var}(A)$ divided by $E(A)$ squared. This illustrates heterogeneity; it is not a correction formula for arbitrary flights. The finite-sample estimate need not vanish even for Gaussian data, and overlapping windows require a matched reference before significance claims.

Velocity memory remains visible after averaging over minutes

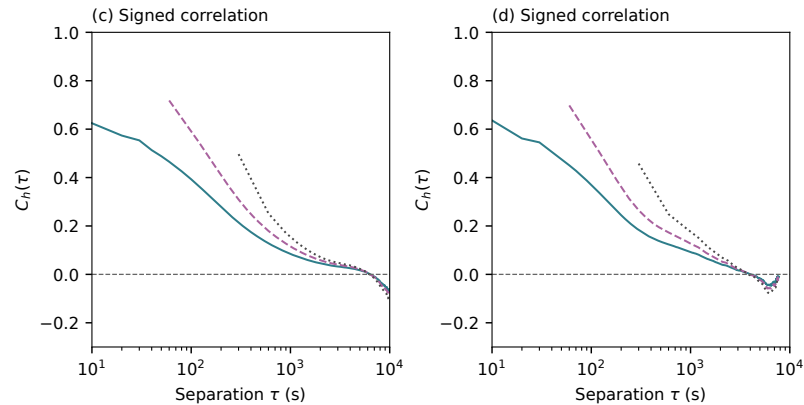
Use $\mathbf{v}_h(t) = [\mathbf{r}(t+h) - \mathbf{r}(t)]/h$ for $h=10, 60, 300$ seconds. Velocities use adjacent non-overlapping blocks; correlations are centred and normalized within flights before averaging. At 600 seconds the para values are 0.128, 0.180 and 0.252. These upper log panels show positive correlations only.



Do the curves support one decay law over the displayed scales?

Signed correlations retain the long-lag negative values

A log plot cannot show the negative part. Both disciplines have long-lag negative values in the signed representation. Coarse averaging can raise normalized correlations without increasing the underlying physical persistence time. At least twenty flights contribute to each plotted estimate.



What share of the long-lag decrease could follow finite-record centring or route returns?

Finite velocity persistence gives a ballistic-to-diffusive crossover

For the same stationary, centred velocity population,

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s) ds.$$

With $C(s) = e^{-s/t_p}$,

$$M_{2,c} = 2\sigma_v^2 t_p [\tau - t_p(1 - e^{-\tau/t_p})].$$

Ballistic at short times, diffusive at long times.

Taylor (1922), pp. 207–208. This covariance identity cannot be applied directly to differently weighted heterogeneous means.

Candidate	Prediction to test	Present conclusion
Homogeneous Brownian	Linear MSD; independent increments	Challenged over the measured range.
Constant drift + Brownian	Drift-free $V_2 \propto \tau$	Simple version challenged; mixtures unresolved.
Homogeneous fBm	Gaussian law; common H_p	Challenged as one pooled law.
Exponential velocity memory	Ballistic–diffusive crossover	Remains a candidate.
Renewal Lévy walk, $1 < \beta < 2$	Two-branch moment spectrum	Finite-record comparison still needed.

“Challenged” means a descriptive mismatch. Calibrated statistical rejection is still to be done.

Each candidate needs its own parameters, environmental assumptions and observation protocol. A long-time diffusive limit has not been tested.

Which models are challenged, and which remain open?

Ordinary Brownian motion is a poor descriptive model of the observed interval, but the asymptotic regime is unknown. fBm is challenged as a homogeneous pooled law; a phase-conditioned or environmentally heterogeneous Gaussian construction is not excluded. The exponential-persistence candidate deserves a joint MSD and VACF comparison. Standard Lévy walk needs finite-record calibration rather than rejection from an illustrative asymptotic line. An unbounded Lévy flight cannot literally describe continuous finite-speed flight; effective truncated versions need separate predictions.

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau | c), \quad w_c = \frac{n_c}{\sum_{c'} n_{c'}}.$$

Here c partitions all admissible windows, including phase histories that cross boundaries. With equal flight weights, apply the identity within each flight before averaging.

The identity holds for any exhaustive labels. Recovering the global curve verifies accounting, not the physical accuracy of those labels.

Which episode durations and between-phase dependencies could distinguish renewal dynamics from persistent motion?

Phase-conditioned moments must include windows crossing transitions

Within-phase windows alone generally omit part of the global moment. Both the mixture weights and conditional moments depend on lag, so global exponents are not weighted averages of phase exponents. A glider moves while climbing; a climb is not automatically a waiting state.

1. **Define the model target.**

Ground-relative motion conditional on environment, or a residual after an independently justified wind correction?

2. **Agree the first calibrated comparisons.**

Fit Brownian controls, fBm, finite persistence and a specified Lévy walk; simulate the same record lengths, selection and estimators.

3. **Choose the most informative next measurements.**

Site-day uncertainty and route matching now; phase durations, increments and between-phase dependence after segmentation validation.

We have measured constraints on growth, shape, geometry and memory. The generating process and an intrinsic Hurst exponent remain unresolved.

Three decisions for this meeting

Agree the physical target, a small set of explicit candidate processes, and a sampling model for uncertainty. No rotation or rescaling has recovered still-air motion. Phase-conditioned comparisons should include transitions and between-episode dependence. Calibrated comparisons can begin before declaring a final physical model.

$$Q_{Y^2}(p) = Q_Y(p)^2, \quad H_{Y^2,p} = 2H_{Y,p} \quad \text{for } Y \geq 0.$$

Squaring bin edges preserves their probability masses. A monotone transform also preserves the supremum distance between the correspondingly transformed CDFs.

If $f_Y(0^+) = c > 0$, then $f_{Y^2}(s) \sim c/(2\sqrt{s})$ near zero. This Jacobian effect says nothing about a heavy upper tail.

Which representation communicates the same result more clearly: displacement magnitude or its square?

Squared laws transform the same distributional evidence

The appendix includes the squared-law figures for both disciplines. Replotting the same observations after a monotone transformation is useful for connecting quantiles with second moments; it does not multiply the amount of independent evidence. Zero mass and dimensional units still need to be carried consistently.

Sources and the scope of the evidence

Zaburdaev, Denisov & Klafter (2015), <i>Lévy walks</i>	Moment-spectrum benchmark
Mandelbrot & Van Ness (1968), <i>Fractional Brownian motions</i>	fBm construction
Taylor (1922), <i>Diffusion by continuous movements</i>	Velocity / displacement relation
Mardia (1970), <i>Measures of multivariate skewness and kurtosis</i>	Centred kurtosis
Efron (1979), <i>Bootstrap methods</i>	Resampling principle

Numerical evidence: current Chapter 3 reports, frozen in the accompanying data and asset manifests. Bibliographic access limitations are retained in the repository audit; citations do not validate the chosen sampling design or constitute calibrated rejection tests.

Joint laws and environmental interpretation: additional sources

Fraiman et al., Cramér–Wold results

Joint laws and projections

Didier & Pipiras, operator fractional Brownian motions

Beyond scalar rescaling

DHV classification guidance

Class is not a performance measure

DREAL Normandie, Le Pays de Caux

Coastal cliffs and valleys

The cited pages support specific mathematical or observational cautions. They do not identify the causes of the regional contrasts measured here.